

The Complex Rung Generator, the State-Angle Chart, and the Residual Sphere

A verified unification of the July 14 explorations
(companion to *The Golden Substrate, the Gate Field, and the Explicit Helix*)

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independent verification (§13):

`ozy_softcheck.py` 91/91 + `relational_softcheck.py` 27/27

`nx_softcheck.py` 27/27 + `qx_softcheck.py` 62/62, exit 0

13 session harnesses (v1.7–v1.8): 349 exact checks + 49 certified guards, all exit 0

Version 1.8 — July 15, 2026

Abstract

Conditional on the whitepaper’s three declared atoms D1–D3, the entire discrete rung ladder is the orbit of a single complex number: the generator $q = i/\varphi = i\tau$, with $W_n = q^n$, $z_n = \sqrt{1 - |W_n|^2}$, and the central identity $q^4 = \varphi^{-4} = \text{gap}$. Golden contraction and the $\mathbb{Z}/4\mathbb{Z}$ quarter-turn become the modulus and phase of one multiplier; the gap first appears as co-intensity after two rungs ($|W_2|^2 = \text{gap}$) and as complex amplitude after one full charge cycle ($W_4 = \text{gap}$). We verify every claim of Ozymandias’ report and of the follow-on thread by four independent harnesses (91 + 27 + 27 + 62 checks: 86+27+24+62 exact decisions plus 5+3 numeric displays, with 3 certified guards alongside the fourth — Findings 12.4, 12.5, 12.6), and unify them with three organizing results: the *state-angle chart* $z = \sin \alpha$, under which the whole gate dictionary becomes elementary trigonometry ($u = \cot^2 \alpha$, $C = \cot^2 \alpha \csc^2 \alpha$, $\sqrt{D} = 1 + 2 \cot^2 \alpha$, $m = -\cos^2 \alpha$, $\lambda = \tan^2 \alpha (1 + \cos^2 \alpha)$), with rung ledgers $(\sin, \cos, \tan)\alpha_1 = (\sqrt{\tau}, \tau, \sqrt{\varphi})$ and $(\sin, \cos, \tan)\alpha_2 = (K, \tau^2, \beta)$ and the general tangent family $\tan \alpha_n = \sqrt{\varphi^{2n} - 1}$; the *K-family theorem* $z_{2j} = \sqrt{1 - \text{gap}^j}$ with the universal two-step law $1 - z_{n+2}^2 = \text{gap} (1 - z_n^2)$ and the uniqueness statement $z_n = K \iff n = 2$; and the *residual-sphere lift* $S(\rho) = (\text{Re } W, \text{Im } W, z) \in S^2$, on which the rungs walk the compass toward the north pole and the whitepaper’s near-miss angle $\theta_K = \arcsin \tau^2$ acquires an exact geometric home as the K-point’s colatitude, $\theta_K = \arccos K$. The residual radius $a(z) = \sqrt{1 - z^2}$ is kept strictly distinct from the inherited formation radius $r(z)$; deriving the latter from the former remains the priority open problem (whitepaper [OPEN] 1). *Version 1.1* adds the Strouse–Ozymandias *canonical quarter-turn factorization*: the imaginary unit of the generator is $J = UA = (H/\sqrt{5})(N/\sqrt{5})$ — the product of the normalized keystone grading and the normalized return-kernel mode exchanger, two real involutions that anticommute *by construction* (the vector kernel of $\frac{1}{2}\{H, \cdot\}$ is exactly $H^\perp$). This part is unconditional keystone algebra, not D1–D3-conditional; it upgrades $Q = \pm \tau J$ (with $Q^4 = \text{gap } I$) from representation-given-declared-*i* to canonical-up-to-orientation, and reduces D2’s declared content to rung *selection* (one Q per floor) rather than the existence of the quarter-turn itself. *Version 1.3* adds the relational bridge: the generator is itself algebraic, with minimal polynomial $x^4 + 3x^2 + 1$, $M = \varphi^2$, and it is exactly the one-charge-quantum gauge rotation of the split golden pair $Z^* = (x^2 - x - 1)(x^2 + x - 1)$; the K-seed sits in the relational-charge catalog with the full $\mathbb{Z}/4\mathbb{Z}$ compass as its angle set; and the transfer-ledger algebra $F_n = z_n^2$, $\Delta F_n = \tau^{2n+1}$, $\sum \tau^{2n+1} = 1$ makes D1 at the rungs a telescoping unit budget. *Version 1.7* adds, verified by eight further session harnesses (234 exact checks + 29 certified guards, all exit 0): the transcendence of the rate $\kappa = \pi/(2 \ln \varphi)$ and of $\log_\varphi 2$ given Gelfond–Schneider, closing the

whitepaper’s near-miss item permanently; the multiplier-torus/systolic *seventh* characterization of the principal branch; the parity no-go for the inherited radius (r^2 is chart-odd: any substrate-to-radius bridge must break $z \mapsto -z$); the universal seed ladder $\text{minpoly}(z_n) = x^4 + c_n x^2 - c_n$ with the Mahler-minimal uniqueness theorem; and the incomplete-elliptic closed form for the above-lens rail with certified non-elementarity. The rate and radius problems remain open as derivations. *Version 1.8* adds, verified by five further session harnesses (115 exact checks + 20 certified guards, all exit 0; running total across the thirteen session harnesses 349 + 49): OP-RATE *closes as a classification* — the discrete-substrate no-go, forcing κ declared-up-to- $\mathbb{Z}$ at rung resolution with the seven characterizations the complete discrete-level menu, each continuum-referencing (the continuum rate D3 stays declared); an apex-ramification valuation certificate and a constructive-bridge obstruction sharpening OP-RADIUS (still open); the χ doctrine’s sub-branches g1/g3 resolved *negatively* on the now-available companion papers (the on-circle image of the emission algebra S is exactly $\mu_2 = \{\pm 1\}$, so $\pm i \notin S$ on-circle, and $W[\text{open}]2$ stays PARTIAL as a derivation); the relational-note cluster (the note’s P4 recovered and reduced to P3, and the odd relational floor bracketed $\mu_S \leq \mu_{\text{rel}}(\text{odd}) \leq 2$ with the $= 2$ promotion left as the corpus’s principal open problem); and the period-relation map ($\tau_0 = \beta + \beta^{-1}$ pinned algebraic, the κ -vs- τ_0 period worry settled negatively, and L_{res} located as an elliptic period at a transcendental parameter).

1 Provenance, scope, and the conditionality banner

Sources. (i) *Report to Ace: Complex Rung Generator and Residual-Sphere Representation* (Ozymandias, 2026-07-14); (ii) the follow-on review thread of the same date (the K-family $z_{2j} = \sqrt{1 - \text{gap}^j}$, the cosine-contraction recursion, and the rung-by-rung trigonometric ledger). (iii) the Strouse–Ozymandias progress report *Toward a Canonical Rung Generator* (2026-07-14): spectral uncoupling, the return-kernel recoupler, and the quarter-turn factorization $J = UA$. (iv) the Strouse–Ozymandias exploratory note *Rail-Pulse Quantization, Helical Path Stretch, and Energy-Gated Recoupling* (2026-07-14) — only its forced kernel is imported here (§11); its rail-pulse mechanism, energy functional, and thermodynamic reading remain hypothesis, as that note itself states. (v) the corpus note *Relational Charge on the Spectral Semiring* (paper 12, 2026-07-01), cited as RC-§, whose machinery this paper’s bridge section uses and whose relevant claims were independently re-derived this session (`relational_softcheck.py`, 27/27). (vi) the second Strouse–Ozymandias exploratory note *Rail-Pulse Propagation, Engine Coupling, and Scale-Dependent Site Density* (2026-07-14) — forced kernel imported (§11); engine-pass and biological layers not imported, per that note’s own claim ceilings. (vii) The whitepaper *The Golden Substrate, the Gate Field, and the Explicit Helix* v1.0 (2026-07-14), whose numbering we cite as W-§. **v1.1 changelog:** §3 added (factorization, coordinate-free form, obstruction lemmas, canonical Q); epistemic ledger and manifest updated; full text of source (i), received post-hoc, integrated (method note above; Remark 6.3). **v1.2 changelog:** §11 added — the forced kernel of source (iv): winding gradient, path-stretch laws, the quarter-turn rail floor $L \geq \pi K/2$ with the resulting one-line divergence proof, and the ν -form of the rung locus with its scoping. **v1.3 changelog:** §4 added (the relational bridge: minimal polynomial of q , the gauge identity with Z^* , the K-seed as catalog object, D2 restated relationally); §11 extended with the site-density trichotomy and the transfer ledger (forced kernel of source (vi)); cross-references and epistemics updated. **v1.4 changelog:** Theorem 2.4 and Corollary 2.5 added — the continuous-interpolation open item is classified (determined up to $\mathbb{Z}$; D3 = principal branch) and merged into the rate problem; Remark 4.5 upgraded with the dilation mechanism. Harnesses now 91 + 27 checks. All other content unchanged. **v1.5 changelog:** Theorem 4.6 added — the complete relational type of the generator object (shells, the cross-shell ν -run, full coherence, twisted-shell realization); §10 extended with Propositions 10.3–10.5 (compass-cross

characterization of the rung locus, the spherical step ledger with the exact quarter-circle first step, the finite spherical length $L_{\text{res}} = \sqrt{1 + \kappa^2} E(\kappa^2/(1 + \kappa^2))$ with the length dichotomy and a fifth characterization extending Corollary 2.5); third harness `nx_softcheck.py` (27 checks) added to §13; Finding 12.3 recorded and the §1 verification paragraph corrected. All other content unchanged. **v1.6 changelog:** §2 extended (Remark 2.6: transfer of the interpolation classification to the formation rail — a sixth characterization of D3); §4 extended (Proposition 4.7: the ratio object at the characteristic level and the Adams square; Theorem 4.8: the quarter twist and the anchor-exchange dichotomy); §8 extended (Theorem 8.3: the K-family is a seed ladder; Corollary 8.4); §11 extended (Proposition 11.7: exact rail-spacing bounds and the gap-rate law, quantifying the divergent side of the length dichotomy of Proposition 10.5); fourth harness `qx_softcheck.py` (62 exact checks + 3 certified guards); Finding 12.4 (front-matter check classification, corrected). Harnesses now 91 + 27 + 27 + 62 checks. All other content unchanged. **v1.7 changelog:** §2 extended — the rate-arithmetic and torus block (Lemma 2.7, Theorem 2.8, Corollary 2.9, Remark 2.10: κ and every branch κ_k transcendental given Gelfond–Schneider; the near-miss item whitepaper [OPEN] 4 closed permanently; W-Thm. 16.1 strengthened to transcendence of $\log_\varphi 2$; the scan-retirement meta-theorem; Proposition 2.11, Theorem 2.12, Remark 2.13: the multiplier torus E_q , its modulus ledger $\mu = \frac{1}{4} + i \ln \varphi / (2\pi)$, the systolic-marking *seventh* characterization of D3, the certified ten-entry geodesic ledger, and the conditional no-CM corollary; Proposition 2.14, Remark 2.15: the forced k -invariance lemma, the exposed-joint axiom H, the transfer-ledger counter-models, and the R6 classification no-go); §3 extended (Propositions 3.7–3.8, Theorem 3.9: pentagon exclusion by the image, unimodular tensor-monomial values ± 1 , the conditional χ -selection chain with the axiom-replacement kill and the de-axiomatization identity); §8 extended (Theorem 8.5: the universal seed ladder; Remark 8.6: the rung-1 Mahler-floor hit, coincidence-candidate; Theorem 8.7 and Remark 8.8: Mahler-minimal uniqueness, the F1 falsifier $x^4 + 6x^2 - 5$, and the register resolution under the F3 reading, signed off 2026-07-15); §9 extended (Proposition 9.2: splitness over $\mathbb{Q}$ is generic, the lens distinguished by its exact data; Remark 9.3: the bounded-negative corpus scan); §10 extended (Theorem 10.6: the RAD-0 parity no-go; Proposition 10.7: the area-transfer law; Remark 10.8: the rotating-equipotential mechanism killed on all three clocks); §11 extended (Theorem 11.8: the incomplete-elliptic closed form for the rail; Corollary 11.9: certified non-elementarity). Eight session harnesses — `tx/mx/rx/ux/ex/cx/lx/dx_softcheck.py` — add 234 exact checks + 29 certified guards, all exit 0 (§13). Findings 12.5 and 12.6 recorded, and the two display counts corrected (`nx`: the 24 + 3 split restored in the §13 sentence; `ozy`: 86 + 5). Register motion: whitepaper [OPEN] 4 closed (contentless, forced given Gelfond–Schneider); the rail closed form resolved (incomplete-elliptic; elementary form impossible, certified); Vein 6 executed; the v1.6 uniqueness question resolved by Theorem 8.7 under the confirmed F3 reading; the rate problem (OP-RATE) remains open as a derivation, with a seventh characterization and a forced k -invariance lemma; the radius problem (OP-RADIUS, whitepaper [OPEN] 1) remains open, sharpened by a forced parity no-go. All other content unchanged. **v1.8 changelog:** §2 extended twice — the OP-RATE classification block (Lemma 2.16: the k -invariance lemma re-forced and extended to the full discrete state; Theorem 2.17: the R6 selection-functional no-go; Corollary 2.18: OP-RATE closes as a classification; Remark 2.19: the seven-way continuum/discrete table), and the period-relation map (Proposition 2.20: τ_0 algebraic and no κ -vs- τ_0 relation over $\mathbb{Q}$; Remark 2.21: L_{res} as an elliptic period and the external-open ledger); §3 extended (Proposition 3.10: the on-circle image of S is μ_2 ; Remark 3.11: the charge character, the g3 dichotomy, and the sharpened $W[\text{open}]2$ verdict); §4 extended (Remark 4.9: P4 recovered and reduced to P3; Proposition 4.10: the bracketed odd relational floor; Remark 4.11: the bounded P1' transport-orbit scan); §10 extended (Theorem 10.9: the apex-ramification valuation certificate; Proposition 10.10: the constructive-bridge obstruction; Remark 10.11: the atom ledger with the honest cap-onset negative). Five session harnesses —

`cl/rd/ch/rn/pm_softcheck.py` — add 115 exact checks + 20 certified guards, all exit 0, bringing the thirteen session harnesses to 349 exact + 49 guards (§13). Register motion: *OP-RATE closes as a classification* (discrete-substrate no-go; κ declared-up-to- $\mathbb{Z}$; the seven characterizations are the complete discrete-level menu, each continuum-referencing; the continuum D3 stays declared, and a continuum-referencing derivation is not excluded — mirroring the $\lambda = 2c/\text{Cencov}$ equivariance no-go); *OP-RADIUS* remains open, sharpened by the apex-ramification certificate (ramification index $e = 2$ for r^2 , $e = 4$ for r) and the constructive-bridge obstruction; $W[\text{open}]2$'s sub-branches $g1/g3$ resolved negatively on the now-available companion papers (on-circle image of $S = \mu_2$, so $\pm i \notin S$), the item staying PARTIAL as a derivation; P4 recovered and reduced to P3; the odd relational floor bracketed (≤ 2 forced, the $= 2$ promotion the corpus's principal open problem); and the period-relation map (τ_0 algebraic, the κ -vs- τ_0 tie settled negatively, L_{res} an elliptic period, and the precise external-open ledger). All other content unchanged.

Method note (from the source report). The investigation began by testing whether K could be read through physical double-helix geometry; the direct DNA-pitch identification *failed quantitatively*. That negative result forced a role classification that shapes everything below: *state space* vs *operator* vs *observable* vs *invariant/landmark*. Under it, the scalar gate iteration $f_C(x) = C/(1+x)$ (W-Decl. 8.1) is the actual operator of the architecture, and K is a rung-2 *landmark* — not the operator, not the space. The generator below is what the operator's rung orbit looks like once contraction and charge are packaged together.

Conditionality banner. Everything in this paper is [FORCED GIVEN D1–D3] — with one important exception: §3's factorization is *unconditional* keystone algebra [FORCED], using no part of D1–D3. Elsewhere the results are exact consequences of the whitepaper's declared chart D1 ($1 - z^2 = e^{-2\rho} = |m|$), charge selection D2, and rate D3 ($d\theta/d\rho = (\pi/2)/\ln \varphi$), together with its forced rung data $\rho_n = n \ln \varphi$, $z_n = \sqrt{1 - \varphi^{-2n}}$, $\theta_n = n\pi/2$ (W-§11–13). Both source documents state this scoping correctly, and this paper preserves it: the representation below *packages* D1–D3; it does not independently derive or justify them. Statements needing only seed arithmetic are [FORCED] outright. The inherited radius profile $r(z)$ is *not* derived here (§10).

Verification. Four independent harnesses (through v1.6): `ozy_softcheck.py`, 86 exact checks + 5 numeric displays (groups OZ-A..OZ-O, of which the OZ-H row is display-only; Finding 12.6); `relational_softcheck.py`, 27 exact checks (RC-A..RC-F); `nx_softcheck.py`, 24 exact checks + 3 numeric displays (NX-R, NX-T, NX-S; NX-H, written cold for v1.5; Finding 12.4); `qx_softcheck.py`, 62 exact checks + 3 certified guards (QX-A..QX-G, written cold for v1.6). All exit 0; sympy 1.14.0; decisions in $\mathbb{Q}/\mathbb{Q}(\sqrt{5})$ extended by i ; K and the K-family decided at the squared level; mpmath display-only except the two interval-certified guard comparisons. Check IDs are cited inline; manifest in §13; the stale count that previously stood in this paragraph is Finding 12.3. v1.7 adds eight session harnesses (`tx`, `mx`, `rx`, `ux`, `ex`, `cx`, `lx`, `dx_softcheck.py`): 234 exact checks + 29 certified guards, all exit 0; manifest and environment pin in §13. v1.8 adds five more session harnesses (`cl`, `rd`, `ch`, `rn`, `pm_softcheck.py`): 115 exact checks + 20 certified guards, all exit 0, read from the Echo-S-Research companion archive read-only and re-encoded cold; running total across the thirteen session harnesses 349 + 49 (§13). Notation as in the whitepaper: $\varphi = (1 + \sqrt{5})/2$, $\tau = \varphi^{-1}$, $\text{gap} = \varphi^{-4}$, $K^2 = 1 - \text{gap}$, $\beta^2 = \varphi^4 - 1 = \varphi^2\sqrt{5}$, $z_c = \sqrt{3}/2$.

2 The generator

Definition 2.1 (Complex residual state). For $\rho \geq 0$ set

$$W(\rho) = e^{-\rho} e^{i\theta(\rho)}, \quad \theta(\rho) = \frac{\pi\rho}{2 \ln \varphi} [\text{D2+D3}],$$

so that

$$W(\rho) = \exp\left[\left(-1 + \frac{i\pi}{2\ln\varphi}\right)\rho\right], \quad |W(\rho)| = e^{-\rho}, \quad \arg W(\rho) = \theta(\rho).$$

Proposition 2.2 (Representation of the chart). [FORCED GIVEN D1–D3] *OZ-G1--G2 D1 is exactly the statement*

$$z(\rho) = \sqrt{1 - |W(\rho)|^2} :$$

residual amplitude decays in the complex plane while height rises as its unit complement. This is a representation theorem for D1–D3, not an independent derivation of them.

Theorem 2.3 (The discrete rung generator). [FORCED GIVEN D1–D3] *OZ-A1--A8, OZ-B1--B3 Let*

$$q = \frac{i}{\varphi} = i\tau.$$

Then at the rungs $\rho_n = n \ln \varphi$,

$$W_n := W(\rho_n) = e^{-n \ln \varphi} e^{in\pi/2} = \varphi^{-n} i^n = (i\tau)^n = q^n, \quad W_{n+1} = q W_n, \quad z_n = \sqrt{1 - |W_n|^2},$$

i.e. every rung simultaneously contracts residual amplitude by τ and rotates the charge by 90° ; the entire discrete ladder is the orbit of the single multiplier q . The exact ledger is $|W_n| = \tau^n$, $\arg W_n = n\pi/2$, $z_n = \sqrt{1 - \tau^{2n}}$. The central identity is

$$q^4 = \left(\frac{i}{\varphi}\right)^4 = \varphi^{-4} = \text{gap},$$

an exact positive real: $i^4 = 1$ restores orientation while τ^4 contracts by the gap. The recurrences

$$W_{n+2} = q^2 W_n = -\tau^2 W_n = -\sqrt{\text{gap}} W_n, \quad W_{n+4} = q^4 W_n = \text{gap} W_n$$

hold exactly (the square root is the positive one: $\tau^2 = \varphi^{-2}$), giving the gap schedule: the gap first appears as co-intensity after two rungs, $|W_2|^2 = \text{gap}$, and as complex amplitude after one complete four-rung charge cycle, $W_4 = \text{gap}$. The architecture does not reset after four rungs: it returns to the same charge direction at residual amplitude reduced by gap and strictly greater height.

Proof. All statements are one-line consequences of $q = i\tau$, $\tau = \varphi^{-1}$, $\tau^2 + \tau = 1$, and $\tau^4 = \varphi^{-4} = \text{gap}$; each is machine-verified exactly for $n = 0.8$ (recurrences), including the sign and branch conventions. The two-rung law is the exact origin of the odd/even rung decomposition (Fibonacci/Lucas branches, W-Thm. 8.1), and the four-rung law is the discrete face of the whitepaper's full-turn cost $\Delta\rho = 4 \ln \varphi = \ln M$ where $M(A_{\text{gap}}) = \varphi^4$ (W-§13): per revolution, co-intensity contracts by $\varphi^{-8} = |W_4|^2$. The imaginary unit written here as i is factored *internally* from the operator layer in §3; the generator is moreover itself an algebraic integer, with minimal polynomial $x^4 + 3x^2 + 1$ and a distinguished place in the relational-charge catalog (§4). $\square$

Theorem 2.4 (Interpolation classification). [FORCED] *OZ-O1--O4 The continuous one-parameter matrix groups through the generator — the curves $G(t) = e^{tL}$ with $G(1) = Q$ — form exactly a $\mathbb{Z}$ -family. Solving $e^{aI+bJ} = Q$ entrywise gives $e^a \cos b = 0$ and $e^a \sin b = \tau$, forcing $\cos b = 0$ and $\sin b = +1$ (the branch $\sin b = -1$ is impossible: $e^a = -\tau$ has no real solution), hence $a = \ln \tau$ uniquely and*

$$\log Q = \left\{ (\ln \tau)I + \left(\frac{\pi}{2} + 2\pi k\right)J : k \in \mathbb{Z} \right\}, \quad W_k(\rho) = e^{-\rho} e^{i(\pi/2 + 2\pi k)\rho / \ln \varphi} \quad (t = \rho / \ln \varphi).$$

All branches agree at every rung, $W_k(n \ln \varphi) = (i\tau)^n$ — the discrete orbit cannot distinguish them — and they are pairwise distinct as curves (initial velocities $-1 + i\kappa_k$ with $\kappa_k = (\pi/2 + 2\pi k)/\ln \varphi$ all distinct; adjacent branches already differ at the half-floor, while branches with $k - k'$ even coincide there but never as functions).

Corollary 2.5 (What D3 selects). [FORCED]/[DECLARED] 0Z-05 The continuous residual state of Definition 2.1 is the $k = 0$ branch, and within the family the following are equivalent: (i) D3's rate; (ii) the principal logarithm of Q ; (iii) minimal winding, $|\pi/2 + 2\pi k|$ minimized; (iv) the shortest residual spiral per floor, $L_k = (1 - \tau)\sqrt{1 + \kappa_k^2}$ minimized (speed $|W'_k| = e^{-\rho}\sqrt{1 + \kappa_k^2}$ integrated over one floor). The interpolation problem is therefore not undetermined but determined up to $\mathbb{Z}$, with D3 exactly the principal-branch selection; deriving that selection from the substrate is the same open problem as deriving the rate.

Remark 2.6 (Transfer to the formation rail). [FORCED GIVEN D1-D3]/[INHERITED] QX-F1--F6 The classification of Theorem 2.4 transfers verbatim to the formation rail: among curves $\Gamma_{\kappa'}(\rho) = (r(z(\rho)) \cos \kappa' \rho, r(z(\rho)) \sin \kappa' \rho, z(\rho))$ whose winding is a one-parameter rotation group and which pass through every rung site, the admissible rates are exactly $\kappa'_k = (\pi/2 + 2\pi k)/\ln \varphi$, $k \in \mathbb{Z}$ — the same $\mathbb{Z}$ -family, because radius and height carry no interpolation freedom (r is inherited, z is D1): the rung-passage condition at $n = 1$ already forces $\kappa' \ln \varphi \equiv \pi/2 \pmod{2\pi}$, and all further rungs follow. All branches agree at every rung site and are pairwise distinct as curves QX-F1--F2; and the $k = 0$ branch is the unique one of shortest *formation-rail* length per floor on both branches of the profile: the speed $\sqrt{r'^2 + r^2 \kappa'^2 + z'^2}$ is strictly increasing in κ'^2 wherever $r > 0$ QX-F3--F5. Corollary 2.5's list thus gains a sixth member — $D3 =$ the shortest formation rail per floor — the rail-side companion of the fifth (shortest total spherical length, Proposition 10.5). The residual-sphere curves $S_k = (\operatorname{Re} W_k, \operatorname{Im} W_k, z)$ inherit the same classification, all on the unit sphere QX-F6.

Rate arithmetic and the multiplier torus (v1.7). The remainder of this section settles the *arithmetic* of the rate $\kappa = \pi/(2 \ln \varphi)$ and packages the branch family of Theorem 2.4 into one geometric object. Scoping first: everything below is *branch-blind* — every statement survives the substitution $\kappa \rightarrow \kappa_k$ — so none of it derives D3. The rate-derivation problem (the session register's OP-RATE) remains open throughout; what changes is its arithmetic perimeter and its classification.

Lemma 2.7 (Irrationality of the log-ratio). [FORCED] TX-A1--A7, TX-G1 $i\pi/\ln \varphi \notin \mathbb{Q}$. If $i\pi/\ln \varphi = p/q$ with $q \geq 1$, exponentiating gives $(-1)^q = \varphi^p$; but $2\varphi^p = L_p + F_p\sqrt{5}$ exactly (exact on the grid $p \in [-12, 12]$, general by Binet) with $\sqrt{5}$ -part $F_p/2 \neq 0$ for $p \neq 0$, so $\varphi^p \neq \pm 1$ for $p \neq 0$ (exact $\mathbb{Q}(\sqrt{5})$ decisions), while $p = 0$ forces $q i\pi = 0$, impossible ($q \geq 1$; $\pi > 3$ interval-certified). Second, premise-independent route: $i\pi/\ln \varphi$ is purely imaginary and nonzero ($\varphi > 1$ exact), hence not rational.

Theorem 2.8 (Transcendence of the rate). [FORCED]/[ESTABLISHED] TX-B1--B5 Forced given the established Gelfond–Schneider theorem (Gelfond 1934; Schneider 1934, independently; quotient form: for nonzero algebraic α, β with fixed logarithm branches and $\log \alpha, \log \beta \neq 0$, the ratio $\log \alpha / \log \beta$ is rational or transcendental) [ESTABLISHED]:

$$\kappa = \frac{\pi}{2 \ln \varphi} \text{ is transcendental, and so is every branch } \kappa_k = (4k + 1) \kappa, \ k \in \mathbb{Z}.$$

Chain: Lemma 2.7 kills the rational horn of G–S at $(\alpha, \beta) = (-1, \varphi)$, so $i\pi/\ln \varphi$ is transcendental; the bridge $i\pi/\ln \varphi = 2i\kappa$ with $2i$ algebraic (root of $x^2 + 4$) transfers transcendence to κ ($\overline{\mathbb{Q}}$ is a field); and $\pi/2 + 2\pi k = (4k + 1)\pi/2$ with $4k + 1$ a nonzero rational for every $k \in \mathbb{Z}$. Restatement-test

outcome, stated explicitly: *this is branch-blind arithmetic — transcendence holds for every branch at once, so nothing here selects $k = 0$; OP-RATE stays open.*

Corollary 2.9 (Near-miss closure). [FORCED]/[ESTABLISHED] TX-C1--C7, TX-G2--G4 $4 \ln \varphi / \pi = 2/\kappa$ exactly, and $2/\kappa$ is transcendental (Theorem 2.8), while τ is algebraic (root of the irreducible $x^2 + x - 1$): the whitepaper’s open item 4 — the near miss $4 \ln \varphi / \pi$ vs τ , margin 5.34×10^{-3} [COMPUTED], there “expected contentless” — closes permanently and negatively. Both clauses of the whitepaper’s Guard 15.3 — the whitepaper carries a single Guard 15.3 with both near-miss clauses; there is no Guard 15.4, which also corrects the citation in Remark 6.5 — promote from [computed-nonequal at 60 dps] to forced-never-equal given G–S: equality would force $\kappa = 2/\tau = 2\varphi$ (root of $x^2 - 2x - 4$) in the first clause, and $\kappa = 1/\tau^2 = \varphi^2$ (root of $x^2 - 3x + 1$) in the pitch clause (via $\tan \theta_K = \tau^2/K$ and injectivity of $\tan$ on $(0, \pi/2)$) — both impossible. General statement: every near miss of the form (algebraic) vs (nonzero rational multiple of $\pi/\ln \varphi$ or $\ln \varphi/\pi$) is permanently empty. Margins interval-certified at 60 dps: $0.00533 < \tau - 4 \ln \varphi / \pi < 0.00534$, $\tau^2 - 2 \ln \varphi / \pi > 0.07$, $1/40 < \kappa - 2\varphi < 3/100$; all printed decimals of Guard 15.3 reproduced TX-G6.

Remark 2.10 ($\log_\varphi 2$ strengthened; scan retirement). [FORCED]/[ESTABLISHED] TX-D1--D5, TX-E1--E6, TX-G5 (a) W-Thm. 16.1 strengthens: $\log_\varphi 2 = \ln 2 / \ln \varphi$ is transcendental given G–S at $(\alpha, \beta) = (2, \varphi)$. Its irrationality is now carried by two premise-disjoint mechanisms, meeting the corpus convergence standard: the whitepaper’s Binet $\sqrt{5}$ -part argument (replicated, grid $p \in [-10, 10]$, $q \in [1, 8]$, both orientations) and the new norm obstruction $|N(\varphi^p)| = |(\varphi\psi)^p| = 1$ vs $N(2^q) = 4^q \geq 4$, independent of Binet positivity. (b) Scan-retirement meta-theorem: $\kappa \notin \overline{\mathbb{Q}}$ (Theorem 2.8), so no algebraic-valued ledger tie for κ exists: for any nonconstant rational function R over $\overline{\mathbb{Q}}$ and any algebraic target c , $R(\kappa) = c$ would make κ algebraic. The register’s empirical no-tie upgrades to a theorem-level exclusion (17 ledger constants certified algebraic with irreducible integer minimal polynomials; κ and $2/\kappa$ retire together, product 2; finite corroboration $|\kappa - c| > 1/40$ over the interval ledger, nearest 2φ at ≈ 0.0282). Honest scope: period relations among the corpus transcendentals (κ , L_{res} , and the register’s τ_0) and the algebraic independence of π and $\ln \varphi$ are beyond Gelfond–Schneider and remain external-open.

Proposition 2.11 (Modulus ledger of the multiplier torus). [FORCED] MX-A1--A4, MX-M1a--M1i, MX-M2a--M2d Let $E_q = \mathbb{C}^\times / q^\mathbb{Z}$ be the multiplier torus of the generator ($0 < |q| = \tau < 1$ exact, so E_q is nondegenerate). The principal logarithm and modulus are

$$\log q = \ln \tau + \frac{i\pi}{2} = -\ln \varphi + \frac{i\pi}{2}, \quad \mu = \frac{\log q}{2\pi i} = \frac{1}{4} + \frac{i \ln \varphi}{2\pi} \pmod{\mathbb{Z}},$$

forced by exp-match and principal-strip membership ($\text{Im} \in (-\pi, \pi]$; branches differ by $2\pi i k$, at least the strip diameter). Ledger: $\text{Re} \mu = \frac{1}{4}$ is the charge quantum — the $c = \frac{1}{4}$ gauge rotation of Theorem 4.2, one charge quantum per rung; $\text{Im} \mu = \ln \varphi / (2\pi) > 0$;

$$\frac{\text{Re} \mu}{\text{Im} \mu} = \kappa \quad \text{and} \quad |\log q|^2 = \ln^2 \varphi + \frac{\pi^2}{4} = \ln^2 \varphi (1 + \kappa^2),$$

the per-floor arc constant of Corollary 2.5(iv) read in the log chart: $(-1 + i\kappa) \ln \varphi = \log q$ exactly.

Theorem 2.12 (Systolic marking: the seventh characterization). [FORCED] MX-M3a--M3f, MX-M4a--M4c For the period lattice $\Lambda = \mathbb{Z} \log q + \mathbb{Z} 2\pi i$ of E_q ,

$$|n \log q + 2\pi i m|^2 = \ln^2 \varphi (n^2 + \kappa^2 (n + 4m)^2)$$

exactly, so every length comparison is linear in κ^2 with integer coefficients, decided by exact rational arithmetic against the single certified sandwich $10 < \kappa^2 < 11$ (one interval guard, *MX-G1*). Λ is branch-independent: $\{\log_k q, 2\pi i\}$ is a unimodular change of basis of Λ for every k , and the length spectrum is k -invariant; the marking (choice of generator) is not $(\log_k q - \log q = 2\pi i k)$. The systole of E_q is $|\log q|$, attained exactly at $\pm \log q$ — the principal-branch generators — hence

D3 $\iff$ the marking whose non-meridian generator is the systole representative.

The second-shortest class is the four-rung cycle: $4 \log q - 2\pi i = \ln \text{gap}$, length $4 \ln \varphi$ — the charge cycle $W_4 = \text{gap}$ of Theorem 2.3 is the second-shortest closed geodesic. Stated prominently: this is characterization number seven for Corollary 2.5’s menu, not a derivation. Per the restatement test the selection content is exactly a minimality principle (the systole); every k -invariant part of the construction cannot separate branches, and OP-RATE’s derivation problem is untouched.

Remark 2.13 (Geodesic ledger; no-CM). [FORCED]/[ESTABLISHED] (a) *MX-M5a--M5c*, *MX-G2--G3* The closed-geodesic length ledger of E_q is certified: lengths $\ln \varphi \sqrt{a + b\kappa^2}$ with $(a, b) = (n^2, (n + 4m)^2)$; the first ten distinct classes (up to sign, iterates included), ordered by exact sandwich arithmetic and interval-corroborated at 60 dps, are

$$(1, 1), (16, 0), (9, 1), (25, 1), (4, 4)^{\times 2}, (49, 1), (64, 0), (36, 4)^{\times 2}, (81, 1), (1, 9),$$

with $(1, 1)$ the systole, $(16, 0) = 4 \ln \varphi$ the $W_4 = \text{gap}$ cycle, and $(4, 4)$ twice the systole (an iterate/primitive degeneracy); window sufficiency ($|n| \leq 12$, $|m| \leq 4$) closed by exact tail identities. This is the certified ledger on which $\pi/2$ and $\ln \varphi$ combine *additively* — the natural habitat for any future extremality selection, which must re-pass the restatement test. (b) *MX-M6a--M6c* Given Theorem 2.8, μ is not algebraic — the exact bridge is $(\mu - \frac{1}{4})\kappa = i/4$ with $\mu - \frac{1}{4} = i \ln \varphi / (2\pi) \neq 0$ — in particular not imaginary quadratic: E_q has no complex multiplication (standard CM criterion; conditional exactly on the transcendence input).

Proposition 2.14 (k -invariance of the discrete state). [FORCED] *DX-E1--E3* The full discrete state of the ladder — the rung values $e^{i\kappa_k n \ln \varphi} = i^n$ (exact on the grid $k \in [-3, 3]$, $n \in [0, 8]$, symbolically via $e^{2\pi i k n} = 1$), the heights $z_n^2 = 1 - \tau^{2n}$, the transfer ledger, the gate data $(C_n, u_n, m_n, \lambda_n)$, the floor lattice $\rho_n = n \ln \varphi$, and the torus lattice Λ — is identical for every branch $k \in \mathbb{Z}$. Only the marking $\log_k q$ is k -dependent (Theorem 2.12).

Remark 2.15 (OP-RATE at rung resolution: sign ladder, exposed joint, ledger kill, classification no-go). [FORCED]/[OPEN] *DX-C1--C7*, *DX-D1--D5*, *DX-E4--E5* Four bookkeeping results at the rate problem’s core; items (i) and (iii) are forced, (ii) and (iv) are candidate-level; none closes the problem. (i) *Sign ladder*, *D1-free* *DX-C1--C2*: symbolically $m(w) = -C/(1 + u)^2 = -1/w < 0$ with $w = \varphi^{2n}$, so the gate multiplier is negative at *every* gate and does not alternate between gates; the two-rung carrier is $q^2 = -\tau^2$, argument π . (ii) *Exposed-joint axiom H* — a formalized open joint, presented as such, *not* a result *DX-C3--C6*: given

$H :=$ “argument of the two-rung ladder action = transverse holonomy accumulated over two floors (= π)”,

exact mod- 2π bookkeeping forces the per-floor step $\delta = \pi/2 \pmod{\pi}$ — the quantum $\pm\pi/2$ at rung resolution — but the residual $\pi\mathbb{Z}$ is not pinned: every branch satisfies H , and the branch remains untouched. (iii) *Ledger counter-models* *DX-D1--D5*: the transfer ledger $\Delta F_n = \tau^{2n+1}$ (Proposition 11.5) does *not* force a constant angular step. Two explicit counter-models pass every ledger identity: $\theta_n = n\pi$ (all moduli correct; fails rung passage at $n = 1$, $e^{i\pi}\tau = -\tau \neq i\tau$) and $\theta_n = n\pi/2 + 2\pi s_n$

with a non-constant integer lift s_n (reproduces *all* rung values i^n and all ledger identities). Constancy of the angular step is therefore a lift/equivariance assumption, and the ledger route to the rate dies by contamination. (iv) *R6 no-go* — a candidate classification statement, not asserted as a theorem **DX-E4--E5**: any rate-selection functional that factors through the k -invariant discrete data of Proposition 2.14 is constant in k and cannot select $k = 0$; the six existing characterizations (Corollary 2.5(i)–(iv), Proposition 10.5, Remark 2.6) all reference *continuum* objects — winding number, spiral speed, rail length — which is exactly the escape, and the separation is exhibited (L_k separates branches that the discrete functionals cannot). This supports closing OP-RATE *as a classification* — determined up to $\mathbb{Z}$ at rung resolution, with selection necessarily continuum-referencing — at candidate level; OP-RATE *as a derivation* remains open **[OPEN]**.

OP-RATE closes as a classification (v1.8). The candidate R6 no-go of Remark 2.15(iv) is now promoted to a forced theorem, and its consequence stated precisely: at rung resolution the branch k — hence the rate κ up to the $\mathbb{Z}$ -family — cannot be selected by any functional of the discrete substrate data. This settles OP-RATE *as a classification*; it is a no-go on discrete-substrate *derivation*, not a derivation of κ , and the continuum rate D3 stays declared throughout.

Lemma 2.16 (Extended k -invariance of the discrete state). **[FORCED]** *CL-1a--CL-1h, CL-1F* The k -invariance of Proposition 2.14 extends, re-forced cold, to the full discrete state of the ladder. For every branch $k \in \mathbb{Z}$ the following are identical to their $k = 0$ values: the rung values $e^{i\kappa_k n \ln \varphi} = i^n$ (exact grid $k \in [-4, 4]$, $n \in [0, 10]$, and symbolically via $e^{2\pi i k n} = 1$); the heights $z_n^2 = 1 - \tau^{2n}$; the gate ladder $(C_n, u_n, m_n, \lambda_n)$ with $m_n \lambda_n = 1$; the floor lattice $\rho_n = n \ln \varphi$; the transfer ledger $\Delta F_n = \tau^{2n+1}$ (sum 1); the torus lattice $\Lambda = \mathbb{Z} \log q + \mathbb{Z} 2\pi i$, re-verified branch-independent by the unimodular change of basis $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ ($\det = 1$); the ten-class geodesic length ledger $\ln \varphi \sqrt{a + b\kappa^2}$ with $(a, b) = (n^2, (n + 4m)^2)$, invariant under the marking relabel $(n, m) \mapsto (n, m + nk)$; and the systole value $|\log q_0|^2 = \ln^2 \varphi (1 + \kappa^2) = \ln^2 \varphi + \pi^2/4$, the lattice minimum. Only the marking $\log_k q$ is k -dependent ($|\log_1 q|^2 - |\log_0 q|^2 = 6\pi^2 \neq 0$, detected *CL-1F*). The lemma consumes neither D2 nor D3 (D1 enters only through the k -free rung modulus τ).

Theorem 2.17 (R6 classification no-go). **[FORCED]** *CL-2a, CL-2F1--CL-2F2* Define a rung-resolution selection functional as any map $\Phi = g \circ D$ from the k -invariant discrete data D of Lemma 2.16 into a totally ordered set, whose argmin/argmax purports to select the branch k . Then every such functional is constant on the $\mathbb{Z}$ -family, and so selects no branch: if $D(k) = D(0)$ for all k (Lemma 2.16), then $\Phi(k) = g(D(k)) = g(D(0)) = \Phi(0)$, whose argmin is the whole $\mathbb{Z}$ -family. (Witness: $F = \sum_{n < 6} e^{-\rho_n} i^n$ is k -constant symbolically *CL-2a*; the non-vacuity guard *CL-2F1* fires on any core secretly made k -constant.) This promotes Remark 2.15(iv) from candidate to forced. Per the restatement test the no-go is branch-blind by design: it is a characterization-of-impossibility, not a derivation of k .

Corollary 2.18 (OP-RATE closes as a classification). **[FORCED]** *CL-3a--CL-3c* OP-RATE closes as a classification. What closes: the reading “derive κ from the discrete substrate at rung resolution” is a no-go (Theorem 2.17) — κ is declared-up-to- $\mathbb{Z}$ at the discrete level, and selecting the principal branch provably requires a continuum-referencing minimality principle. The seven characterizations (Corollary 2.5(i)–(iv), Proposition 10.5, Remark 2.6, Theorem 2.12) are the complete discrete-level menu: the discrete level contains no selector at all. What does not close: the continuum rate D3 itself remains declared — all branch curves W_k are admissible one-parameter groups through Q , pairwise distinct as curves (Theorem 2.4, initial velocities $-1 + i\kappa_k$) — and a continuum-referencing derivation is not excluded. This mirrors the $\lambda = 2c/\text{Cencov}$ equivariance no-go of the companion $\lambda = 2c$ paper (2026-06-lambda-2c, thm:cencov) exactly: as there “ c cannot be fixed by any invariance requirement ... c is a declared positive scale ... shipped decisions are

c-invariant across the band ... [but he] does not forbid an external structural constraint from doing so”, here k cannot be fixed by any functional of the k -invariant discrete data, κ is declared-up-to- $\mathbb{Z}$ with discrete decisions k -invariant across the $\mathbb{Z}$ -family, and a continuum-referencing principle may still select. *OP-RATE* as a derivation remains open [OPEN].

Remark 2.19 (The seven-way continuum/discrete table). [FORCED] CL-2b(i)--(vii), CL-2c The content of the no-go is that each of the seven characterizations is *continuum-referencing*: its selection functional is k -dependent (unique argmin at $k = 0$), so it provably does not factor through the k -invariant discrete data. (i) D3’s declared continuum rate κ itself — a *restatement* (the selector $|\kappa_k - \kappa| = 4|k|\kappa$ consumes the target); (ii) the principal branch of $\log Q$, i.e. the continuous argument/lift $\theta(\rho)$ (the rung data is lift-blind, $e^{i\theta} = i^n$ only); (iii) the minimal winding $|\pi/2 + 2\pi k|$ of the continuous curve; (iv) the arc length $L_k = (1 - \tau)\sqrt{1 + \kappa_k^2}$ of the continuous residual spiral; (v) the total spherical length L_{res} ; (vi) the formation-rail length per floor; (vii) the systolic selection — the flat norm $|\log_k q|^2 = \ln^2 \varphi(1 + (4k + 1)^2 \kappa^2)$ of the k -dependent *marking* on the continuous torus E_q , even though the lattice and the systole value are k -invariant (Lemma 2.16). Tag discipline: the structural conclusion “continuum-referencing, does not factor through the discrete data” is [FORCED] for all seven, and rows (i)–(iv),(vii) are forced at the exact level (integer/rational cores $(4k + 1)^2$, $|4k + 1|$, $4|k|$, plus $\kappa^2 > 0$); the strict per-characterization “unique argmin at $k = 0$ ” *selection* for rows (v) and (vi) is [FORCED] *given corpus* — inherited, cited not re-encoded, from Proposition 10.5 (L_{res} strictly increasing in $|\kappa|$) and Remark 2.6 (rail speed strictly increasing in κ'^2). The no-go’s core needs only that L_{res} and the rail are continuous integrals, manifestly not functions of the rung samples, which is exact. The whole block is branch-blind by design: that is the content.

The period-relation map (v1.8). The scan-retirement remark (Remark 2.10(b)) left the period relations among the corpus transcendentals (κ , L_{res} , and the register’s Salem-slot constant τ_0) external-open and undifferentiated. This block pins what is algebraic, settles the κ -vs- τ_0 worry, and maps the precisely external-open residue.

Proposition 2.20 (τ_0 algebraic; no κ -vs- τ_0 relation). [FORCED]/[ESTABLISHED] PM-1B1--PM-1B9, PM-1X1--PM-1X6, PM-A2 The Salem-slot occupant $\tau_0 = \beta + \beta^{-1}$ (the trace redirection of the companion salem-slot note, Def. 3.2; the dominant root of the degree-2 trace-down) is algebraic. For the least-degree Salem number β_4 (minimal polynomial $x^4 - x^3 - x^2 - x + 1$) it has the exact minimal polynomial

$$\boxed{\text{minpoly}(\tau_0) = t^2 - t - 3, \quad \tau_0 = \frac{1}{2}(1 + \sqrt{13})},$$

a degree-2, irreducible, monic-integer (algebraic-integer), totally real (discriminant $13 > 0$, Sturm real-root count 2), Perron ($\tau_0 - |\text{conjugate}| = 1 > 0$) constant with $\tau_0 > 2 > \varphi$ (salem-slot Thm. 3.3); the type-fact is general over all Salem redirections, and the golden limit $\sqrt{5} = \varphi + \varphi^{-1}$ (minpoly $t^2 - 5$) and the Pisot accumulation $\mu_P + \mu_P^{-1}$ (minpoly $t^3 + t^2 - 4t - 5$) are algebraic too. Combined with κ transcendental (inherited, Theorem 2.8, given Gelfond–Schneider), there is no algebraic relation over $\mathbb{Q}$ between κ and τ_0 : for $m(Y) = \text{minpoly}(\tau_0)$ the resultant $R(X) = \text{Res}_Y(P, m) = \prod_r P(X, r) \in \mathbb{Q}[X]$, and a genuine (non-vacuous) $P(\kappa, \tau_0) = 0$ would give $R(\kappa) = 0$ with $R \neq 0$, forcing $\kappa \in \overline{\mathbb{Q}}$ — impossible. The register’s “ κ vs τ_0 ” period worry is thus settled negatively at the algebraic level (a specialization of the scan-retirement meta-theorem to two variables; κ is consumed only as an excluded value, not derived, so this is branch-blind exclusion, not restatement).

Remark 2.21 (L_{res} as an elliptic period; the external-open ledger). [FORCED]/[ESTABLISHED]/[OPEN] PM-1C1--PM-1C2, PM-2A, PM-2B, PM-2Bf The finite spherical length $L_{\text{res}} = \sqrt{1 + \kappa^2} E(\kappa^2/(1 + \kappa^2)) =$

3.7312193125... (Proposition 10.5) is the value of the complete elliptic integral of the second kind at the elliptic parameter $m = \kappa^2/(1 + \kappa^2)$ (modulus $k = \sqrt{m}$), via the exact reduction $1 + \kappa^2 \cos^2 \theta = (1 + \kappa^2)(1 - m \sin^2 \theta)$. At an algebraic parameter E is a Kontsevich–Zagier period (an algebraic ellipse’s quarter-perimeter); here the Möbius inverse $\kappa^2 = m/(1 - m)$ makes m transcendental (since κ is), so L_{res} is a period function evaluated at a transcendental argument — one step beyond Gelfond–Schneider and beyond the Schneider/Chudnovsky elliptic-period theorems (which need an algebraic modulus or CM lattice), hence its transcendence is *not* settled here. Since τ_0 is algebraic (Proposition 2.20), any relation $P(\tau_0, L_{\text{res}}) = 0$ over $\mathbb{Q}$ would force $L_{\text{res}} \in \overline{\mathbb{Q}}$ by the same resultant/tower argument PM-2B, so the (τ_0, L_{res}) pair *reduces* to the single constant. The precise ledger: *settled* — κ transcendental, τ_0 algebraic, no (κ, τ_0) relation, and (τ_0, L_{res}) reduced to “is L_{res} transcendental”; *genuinely external-open* [OPEN] — the transcendence of L_{res} , and the algebraic independences of (κ, L_{res}) and of $(\pi, \ln \varphi)$ (Schanuel / Kontsevich–Zagier territory, beyond Gelfond–Schneider and Baker; the ratio $\pi/\ln \varphi = 2\kappa$ is settled transcendental but pair-independence is strictly stronger). The entire map survives $\kappa \mapsto \kappa_k$ (branch-blind) and advances no selection.

3 The canonical quarter-turn: $J = UA$

Everything in this section is unconditional [FORCED]: pure keystone/Clifford algebra in the whitepaper’s fixed convention ($R = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, $H = 2R - I = 2e_1 - e_2$, $e_1 = \sigma_x$, $e_2 = \sigma_z$, $J = e_1 e_2$; W-§3–4), using no part of D1–D3. It resolves the canonicity problem the July 14 report posed for itself: the representation $W_n = (i\tau)^n$ imported i from the declared winding; here the quarter-turn is *factored from two real, normalized structures already forced in the operator layer*.

Proposition 3.1 (Spectral uncoupling). [FORCED] OZ-J1--J5 *The spectral projectors of the keystone,*

$$P_+ = \frac{R + \tau I}{\sqrt{5}}, \quad P_- = \frac{\varphi I - R}{\sqrt{5}},$$

are complete orthogonal idempotents ($P_+ + P_- = I$, $P_\pm^2 = P_\pm$, $P_+ P_- = 0$) with $RP_+ = \varphi P_+$ and $RP_- = \psi P_-$, $\psi = -\tau$. The normalized keystone grading

$$U := P_+ - P_- = \frac{H}{\sqrt{5}} = \frac{2R - I}{\sqrt{5}}$$

satisfies $U^2 = I$ and acts as $+I$ on the expanding golden mode and $-I$ on the contracting one; equivalently $P_\pm = (I \pm U)/2$.

Proposition 3.2 (The return-kernel recoupler). [FORCED] OZ-K1--K4, OZ-L4 *Let $N = e_1 + 2e_2$ be the generator of the vector part of $\ker \mathcal{L}$, $\mathcal{L}(X) = \frac{1}{2}\{H, X\}$ (W-Thm. 4.2). Then $N^2 = 5I$ and, normalizing by the Clifford norm, $A := N/\sqrt{N^2} = N/\sqrt{5}$ satisfies*

$$A^2 = I, \quad UA = -AU, \quad AP_+ = P_- A, \quad AP_- = P_+ A :$$

A is a real involution that exchanges the two golden modes. The anticommutation is automatic, not a coincidence: on vectors, $\mathcal{L}(v) = \langle H, v \rangle I$ (W-Thm. 4.2), so the vector kernel is by definition $H^\perp$ — membership in the return kernel already certifies orthogonality to H , hence $HN = -NH$.

Theorem 3.3 (Canonical quarter-turn factorization; Strouse–Ozymandias). [FORCED] OZ-K5--K6
 With U and A as above,

$$\boxed{J = UA = \frac{H}{\sqrt{5}} \cdot \frac{N}{\sqrt{5}} = \frac{HN}{5}}, \quad J^2 = UAU A = -U^2 A^2 = -I.$$

Matrix witness: $H = \begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}$, $N = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$, $HN = 5 \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = 5e_1e_2$. *Uniqueness:* U is fixed by the keystone; the vector kernel is one-dimensional, so its unit generator is $\pm A$; hence

$$J = \pm UA$$

is canonical up to the orientation sign — exactly the freedom the whitepaper’s D2 already names as a convention. The quarter-turn no longer needs to be imported: distinguish the modes $(U) +$ exchange the modes $(A) =$ quarter-turn (J) .

Corollary 3.4 (Coordinate-free form). [FORCED] OZ-L3--L4 In $\text{Cl}(2, 0)$ the product of any positively-oriented orthonormal pair of vectors equals the pseudoscalar: for every rotation angle θ , $e'_1e'_2 = e_1e_2 = J$ exactly (a reflection flips the sign). Since $(U, A) = (\hat{H}, \hat{N})$ is an orthonormal pair by construction (Prop. 3.2), the factorization $J = \pm UA$ is basis-independent up to orientation; no choice beyond the whitepaper’s fixed convention enters. For an arbitrarily scaled kernel generator the normalization is $A = N/\sqrt{N^2}$; with $N = e_1 + 2e_2$ this is $N/\sqrt{5}$. This closes the report’s open item (1).

Remark 3.5 (Why the return layer is necessary). [FORCED] OZ-L1--L2 The commutative algebra $\mathbb{R}[R] = \text{span}\{I, R\}$ contains no square root of $-I$: $(aI + bR)^2 = -I$ has no real solutions (exact elimination). Nor does it contain the recoupler: everything in $\mathbb{R}[R]$ commutes with H , while $[H, N] = 2HN = 10J \neq 0$. The quarter-turn therefore cannot come from the keystone *alone*; it requires the keystone *acting on* the matrix algebra — the return operator’s module structure — which is precisely where N lives. The report’s initial self-criticism (“clean representation $\neq$ canonical derivation”) identified the right gap, and the factorization fills it from inside the existing layer.

Definition 3.6 (Canonical rung generator). [FORCED] OZ-K7--K9

$$Q := \pm \tau J = \pm \tau UA = \pm \frac{\tau}{5} HN, \quad Q^2 = -\tau^2 I, \quad Q^4 = \text{gap } I, \quad Q^n = \tau^n J^n.$$

Under the algebra isomorphism $\mathbb{R}[J] \cong \mathbb{C} (J \mapsto i)$, $Q^n \cong (i\tau)^n = W_n$; concretely, with basepoint $v_0 = (1, 0)^\top$,

$$Q^n v_0 = (\text{Re } W_n, \text{Im } W_n)^\top, \quad \text{so} \quad S_n = (Q^n v_0; z_n) :$$

the residual sphere’s horizontal part (§10) is the Q -orbit of the east basepoint.

Scoping upgrade (what this does and does not close). Before v1.1 the generator was forced *given* D1–D3, with i supplied by D2. Now: the quarter-turn operator J and the rung generator $Q = \pm \tau J$ exist canonically (up to orientation) in the substrate itself [FORCED], and D2’s remaining declared content is *selection*, not existence — that the winding applies exactly one Q per entropy floor (D3’s rate), rather than the terrain π or pentagon $2\pi/5$ clocks. Per the report’s own ledger, still open: rung quantization (why exactly one Q per rung event — by Corollary 2.5 this now *contains* the interpolation question, since branch selection = rate selection), derivation of the inherited radius law $r(z)$, and any physical identification (the helicase/uncouple–recouple language

was a mechanism-finding analogy only). Coordinate-free uniqueness is closed by Corollary 3.4; continuous interpolation is classified by Theorem 2.4.

The χ -selection front (v1.7). Three results sharpen the selection scoping just stated: what the emission image S (the closure of the whitepaper’s emission catalog, W-§5) can and cannot supply for the angular quantum χ , and exactly where a χ -selection argument must stand.

Proposition 3.7 (Pentagon exclusion by the image). [FORCED] CX-A3, CX-B1--B5, CX-D6 ζ_5 is not in the emission image S : $\arg \zeta_5 = 2\pi/5$ lies outside the image angle lattice $(\pi/2)\mathbb{Z}$ (W-Lemmas 5.4–5.5). Exactly: $\frac{1}{5} \notin \frac{1}{4}\mathbb{Z} \bmod 1$; the pentagon classes $k/5$ ($k = 1..4$) meet $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ only at 0; Φ_5 (verified = $\minpoly(\zeta_5)$) has no root in μ_4 ; and the pentagon gate multipliers are primitive fifth roots (pair product = Φ_5). This strengthens the whitepaper’s pentagon exclusion from the principle to the image: W-Decl. 12.2/W-Prop. 14.2 exclude the $2\pi/5$ clock by the principle D2; here the pentagon multiplier is not an on-circle element of S at all, independently of D2.

Proposition 3.8 (Unimodular tensor-monomials reach only ± 1). [FORCED] CX-A4, CX-C1--C7 Every unimodular tensor-monomial of catalog eigenvalues has value exactly $+1$ or -1 : $\pm i$ is multiplicatively unreachable. Forcing: every catalog eigenvalue has 5-valuation $w_2 \geq 0$ (moduli of the form $2^{w_0/4}3^{w_1/4}5^{w_2/4}\varphi^{w_3/4}$); the only odd-quarter-argument carriers are the K -seed’s rotation pair $\pm i\beta$, each with $w_2 = 1 > 0$; unimodularity forces the total $w_2 = 0$ (norming to $\mathbb{Q}$ with $N(\varphi) = -1$ and φ not a root of unity), hence no $\pm i\beta$ factor, hence even argument. Corroborated exhaustively on all 74613 monomials of degree ≤ 6 (60 unimodular, all ± 1). Whether the full image — closed also under addition, minimal polynomials, and Φ -extraction — can reach $\pm i$ is not decided by this multiplicative argument and remains open [OPEN] CX-C6--C7: the corpus-exhibited host is $x^4 - 1 = \Phi_1\Phi_2\Phi_4$ with Φ_4 rooted at $\pm i$.

Theorem 3.9 (The conditional χ -selection chain, with its kill). [FORCED]/[DECLARED] CX-D2--D3, CX-E1--E5, CX-G1 Call $D2'$ the thin axiom “the per-rung winding multiplier is an on-circle element of S ”. Then, under $D2'$ together with the hypothesis that S contains a non-real on-circle element: $\chi = \pm\pi/2$. The chain composes three links, each re-verified verbatim this session: on-circle elements of S lie in $\mu_4 = \{\pm 1, \pm i\}$ (W-Lem. 5.6); real multipliers carry no helicity (W-Thm. 11.1/W-Cor. 11.2), excluding ± 1 ; leaving $\chi = \pm\pi/2$. The implication is machine-forced as a truth table over the hypothesis. The honest kill, stated prominently: $D2'$ does not reduce D2. Selecting $\pm i$ requires S to contain $\pm i$, and the substrate’s only odd-quarter-argument source is the K -seed’s rotation pair $\pm i\beta$ — so motivating “ $D2'$ puts $\pm i$ in S ” imports exactly D2’s declared content (the winding realizes the seed’s charge completion, Remark 4.4): axiom replacement is a restatement, and the kill is logged. No shipped document ties the forced gate multiplier $\pm i$ (a $C = -\frac{1}{2}$ gate datum, W-Prop. 10.2) to the emission image, so $D2'$ remains a proposed thin axiom, not forced-given-corpus CX-D3, CX-D6. What survives unconditionally is the de-axiomatization identity CX-E3: within μ_4 , non-real $\iff$ order 4 $\iff$ generator of $\mathbb{Z}/4\mathbb{Z}$ — so D2’s clause “the angular quantum is a generator of $\mathbb{Z}/4\mathbb{Z}$ ” may be rewritten as “the multiplier is non-real” without weakening it. D2 itself remains declared [DECLARED].

The χ doctrine resolved (v1.8). Theorem 3.9 left two sub-branches open pending companion papers: g1 (can the full emission image S — closed also under addition, minimal polynomials, and Φ -extraction — reach $\pm i$?), and g3 (does the operator-algebra / charge-measure doctrine require dynamical gate multipliers to be elements of S ?). Those papers are now available and re-encoded cold; both sub-branches resolve *negatively*, and W[open]2 stays PARTIAL as a derivation.

Proposition 3.10 (The on-circle image of S is μ_2). [FORCED]/[ESTABLISHED] CH-A3, CH-B2--CH-B3, CH-C1--CH-C5, CH-D2, CH-G2 Let S be the emission algebra — the closure of the whitepaper’s 7-seed

emission catalog under $\{\oplus, \otimes, \psi^2, \text{minpoly}, \Phi\}$ (commutator excluded). Its on-circle eigenvalue image is exactly $\mu_2 = \{+1, -1\}$, a proper subset of μ_4 ; hence $\pm i$ is not an on-circle element of S . Mechanism (forced given the operator-algebra corpus): every emitted eigenvalue is a monomial in the 16 catalog eigenvalues — $\otimes$ multiplies, ψ^2 squares, $\oplus$ adds no value, and minpoly, Φ are spectrum-preserving idempotents (operator-algebra paper, 2026-06-operator-algebra: they re-present an object's eigenvalue multiset, they are not a cyclotomic-extraction primitive) — and a unimodular monomial forces total 5-valuation 0 ($N(\varphi) = -1$, φ not a root of unity), which excludes the only odd-argument carriers $\pm i\beta$ (with 5-valuation 1), forcing even argument, hence value in $\{\pm 1\}$. This resolves g1 (v1.7 external-open) negatively: the Φ -extraction host $x^4 - 1 = \Phi_1\Phi_2\Phi_4$ is not in S because spectrum-preservation cannot manufacture the $\pm i$ roots of Φ_4 , and the emission-gap paper's on-circle lemma (2026-06-emission-gap: an emitted on-circle eigenvalue lies in $\{1, i, -1, -i\}$) is a containment (an upper bound, image $\subseteq \mu_4$), not the equality $\mu_4 \subseteq S$. This also reconciles with the tensor-monomial sub-closure of Proposition 3.8 (which reached only ± 1): there is no gap on-circle between the sub-closure and the full image, since ψ^2 preserves the on/off-circle dichotomy and no catalog seed has $\pm i$ as a root. The only argument- $\pi/2$ eigenvalues anywhere in S are the off-circle $\pm i\beta$ at modulus $\beta = 5^{1/4}\varphi = 2.4195\dots > 1$ CH-G2.

Remark 3.11 (The charge character, g3, and the sharpened $W[\text{open}]2$ verdict). [FORCED]/[ESTABLISHED] CH-D1--CH-D3, CH-E1--CH-E6, CH-G1 The *charge* character does contain the non-real charges, but this does not force $\chi = \pm\pi/2$. $\chi(S) = \mathbb{Z}/4\mathbb{Z}$ carries the odd charge classes 1, 3 (operator-algebra paper: the K -seed carries $\{0, 1, 2, 3\}$, its imaginary pair $\pm i\beta$ at $\pm\pi/2$); but charge reads the *argument* class only (modulus-free), and those charge-1, 3 elements are the off-circle $\pm i\beta$ — so the hypothesis of Theorem 3.9 that “ S contains a non-real on-circle element” is *false given corpus* (Proposition 3.10): the non-real content of S lives entirely off the circle. Hence g3 resolves negatively — the doctrine does not require dynamical gate multipliers to be on-circle elements of S — as a truth-table over the thin axiom D2': Reading A (“the multiplier is an on-circle element of S ”) is *unsatisfiable* for the rotation gate, whose forced multiplier is $\pm i$ (a $C = -\frac{1}{2}$ gate datum, W-Prop. 10.2) while on-circle $S = \{\pm 1\}$ and helicity removes the reals; Reading B (“the multiplier's charge lies in $\chi(S)$ ”) exists only via the K -seed rotation pair, so it *imports exactly D2* (the charge completion) — a restatement. The charge-measure-coupling paper independently notes that the matrix commutator is the one non-semiring door across which emission constraints do not extend. *Net*: $W[\text{open}]2$ does *not* upgrade to a $\chi = \pm\pi/2$ derivation — it stays PARTIAL as a derivation, now sharpened and grounded, its openness precisely located at the unreduced axiom D2. The two v1.7 opens are closed as *forced-given-corpus negatives* (g1: external-open $\rightarrow \pm i \notin S$ on-circle; g3: papers-absent no-tie $\rightarrow$ the D2' dichotomy above). Kills logged: D2'-as-a-doctrine-theorem under both readings, and the $\mu_4 \subseteq S$ misreading of the emission-gap containment. Any χ -selection remains a branch-blind *characterization* of the quantum ($\pi/2 \bmod 2\pi$ for all k), never a branch derivation CH-E6.

4 The generator as a relational object

The relational-charge note (source (v)) attaches phase data to *objects* — conjugation-closed root multisets of monic integer polynomials — through the angle $t(\alpha) = \frac{1}{2\pi} \text{Arg } \alpha$, the relational group Δ generated by pairwise differences, the absolute anchor $n \in \{m, 2m\}$ (its Rigidity Theorem RC-4.1), and the complete cyclotomic contact signature of the ratio object Rat_p . Its own guardrail applies throughout: the vocabulary is structural, not physical. This section proves that the objects of the present paper live in that catalog, and that the generator itself is one of its distinguished witnesses. Everything here is unconditional [FORCED], machine-verified in the second harness (RC-D3, RC-E1--E7).

Theorem 4.1 (Minimal polynomial of the generator). [FORCED] RC-E3, RC-E5 The rung generator $q = i\tau$ is an algebraic integer with minimal polynomial

$$\boxed{\text{minpoly}(q) = x^4 + 3x^2 + 1}, \quad \text{roots } \{\pm i\tau, \pm i\varphi\}, \quad M = \varphi^2,$$

a reciprocal unit quartic ($p(0) = 1$), irreducible over $\mathbb{Q}$. Its angle set is $\{\frac{1}{4}, \frac{3}{4}\}$: absolute charge group $\mathbb{Z}/4\mathbb{Z}$ with charges $\{1, 3\}$ (odd charges only), while the pairwise differences generate only $\Delta = \mathbb{Z}/2\mathbb{Z}$ — a group-drop object, the same coherence type as the note's canonical drop witness (its ledger entry E , $x^4 + 5x^2 + 5$), with contact signature $\{\Phi_1^4, \Phi_2^4\}$.

Proof. $q^2 = -\tau^2$, so $2q^2 + 3 = \sqrt{5}$ and squaring gives $q^4 + 3q^2 + 1 = 0$; the roots are $\pm i\tau, \pm i\varphi$ since $x^2 \in \{-\tau^2, -\varphi^2\}$, verified by substitution. Irreducibility, the Mahler value $\varphi \cdot \varphi = \varphi^2$ (the two roots of modulus φ), and the complete contact scan are machine checks. $\square$

Theorem 4.2 (Quarter-turn gauge identity). [FORCED] RC-D3, RC-E4, RC-E5 Let $Z^* = (x^2 - x - 1)(x^2 + x - 1) = x^4 - 3x^2 + 1$, the split golden pair (roots $\{\pm\varphi, \pm\tau\}$) — the relational note's hypothesis-necessity witness for its modulus-pinning theorem. Then, in both directions,

$$\boxed{\text{minpoly}(q)(ix) = Z^*(x) \quad \text{and} \quad Z^*(ix) = \text{minpoly}(q)(x)} :$$

the generator's object is exactly the $c = \frac{1}{4}$ gauge rotation — one charge quantum — of the split golden pair. Consequences:

- (a) **Gauge-blindness realized:** Rat is invariant under the rotation (a global root scaling cancels in every ratio), so both objects carry the identical signature $\{\Phi_1^4, \Phi_2^4\}$ — a live instance of the note's Proposition RC-3.5(iv) (rotation fixes every relative charge) and of its gauge-blind probe (RC-Rem. 7.6).
- (b) **Both rigidity anchors from one pair:** Z^* has $(m, n) = (2, 2)$ (all roots real, angles $\{0, \frac{1}{2}\}$) while iZ^* has $(m, n) = (2, 4)$ (angles $\{\frac{1}{4}, \frac{3}{4}\}$). The anchor dichotomy $n \in \{m, 2m\}$ of Rigidity is realized by a single gauge pair, exchanged by exactly the seed's charge quantum.
- (c) The torsion ratio -1 at the shared modulus φ inside Z^* — the very relation that makes Z^* the pinning witness — is the golden internal relation $t_{\text{rel}}(\varphi, \varphi') = \frac{1}{2}$ (RC-Lem. 6.1) seen across the two factors.

Proposition 4.3 (The K-seed is a catalog object). [FORCED] RC-E1 The K-seed $x^4 + 5x^2 - 5$ (W-§2) has roots $\{\pm K, \pm i\beta\}$ at angles $\{0, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}\}$ — **the full charge compass** — with $\Delta = \mathbb{Z}/4\mathbb{Z}$, anchor $n = m = 4$, $M = \beta^2$ (the whitepaper constant), and contact signature $\{\Phi_1^4, \Phi_2^4\}$: it is the relational note's ledger entry F , whose sign-mirror is the drop witness of Theorem 4.1. The two research lines share this object at the root level.

Remark 4.4 (D2, restated relationally). [FORCED]/[DECLARED] RC-E2 The winding's rung angles $\{n/4\}$ have relational group $\mathbb{Z}/4\mathbb{Z} = \Delta(\text{K-seed})$: Declaration D2 — the winding realizes the seed's charge completion — has the reference-free form the helix's angle lattice realizes the K-seed's relational charge group. This is an identification at the angle-lattice level, gauge-invariant by construction; D2 itself remains declared. Likewise the ν -locus of Proposition 11.3 now stands directly on the note's real-pair lemma: $\nu(\rho_n) = (-1)^n$ is torsion of order ≤ 2 , so the rungs are precisely the real-coherence events of the relational ν -criterion RC-E7.

Remark 4.5 (Mahler bookkeeping across the bridge — with mechanism). [FORCED] RC-E6, OZ-06 $K = 5^{1/4}\tau$ and $\beta = 5^{1/4}\varphi$ exactly (from $(K\varphi)^4 = 5$ and $\beta = K\varphi^2$), so the K-seed's root moduli

$\{K, K, \beta, \beta\}$ are the $5^{1/4}$ -dilation of the generator object's root moduli $\{\tau, \tau, \varphi, \varphi\}$. The dilation carries exactly the two β -roots outside the unit circle, multiplying the measure by $(5^{1/4})^2$:

$$M(K\text{-seed}) = \beta^2 = \sqrt{5} \varphi^2 = \sqrt{5} \cdot M(\text{minpoly}(q)).$$

The bridge factor $\sqrt{5}$ is the dilation's Mahler cost.

Theorem 4.6 (Complete relational type of the generator object). [FORCED] NX-T1--T9 *The root multiset of $\text{minpoly}(q) = x^4 + 3x^2 + 1$ splits into exactly two shells (equal-modulus classes, RC-Def. 7.1): $\{\pm i\tau\}$ at modulus τ and $\{\pm i\varphi\}$ at modulus φ — conjugate pairs on the imaginary axis, exchanged by root inversion (the quartic is reciprocal). The complete ordered-pair ledger over all 16 root pairs is*

$$\underbrace{1^{\times 4}}_{\Phi_1^4 \text{ (diagonal)}}, \quad \underbrace{(-1)^{\times 4}}_{\Phi_2^4 \text{ (antipodal)}}, \quad \underbrace{\pm\tau^2, \pm\varphi^2 \text{ (each } \times 2)}_{\text{cross-shell: all real, none unimodular}},$$

so the contact signature $\{\Phi_1^4, \Phi_2^4\}$ of Theorem 4.1 is complete: the only unimodular ratios are ± 1 , and no further cyclotomic contact exists (complete scan bound $M \leq 2 \cdot 16^2 = 512$, RC-Lem. 7.4). Cross-shell coherence, decided by the ν -criterion (RC-Rem. 7.6: $\nu = \mu/\bar{\mu}$ for $\mu = \alpha/\beta$), holds identically: every cross ratio is real, so $\nu \equiv 1$ on all eight cross-shell pairs, with relative charge $t_{\text{rel}} = 0$ on equal-angle pairs and $t_{\text{rel}} = \frac{1}{2}$ on opposite-angle pairs. The generator object is therefore fully coherent — a single coherence class spanning both shells, fixed by the negation involution — with complete type

$$\text{angles } \{\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \frac{3}{4}\}, \quad \Delta = \mathbb{Z}/2\mathbb{Z}, \quad (m, n) = (2, 4), \quad M = \varphi^2, \quad \{\Phi_1^4, \Phi_2^4\}.$$

Structurally it is a twisted shell in the relational note's sense (its open problem (P1'); twisted shells are its Example 7.19 mechanism): $x^4 + 3x^2 + 1 = s(x^2)$ with $s(y) = y^2 + 3y + 1 = \text{minpoly}(q^2)$ — the $k = 2$ twist of the real quadratic of $q^2 = -\tau^2$ — realized here on the admissible side, where full coherence is automatic (RC-Cor. 4.2) but the shell, contact, and ν data remain contentful. The cross-shell coherence witnesses have moduli exactly $\{\tau^2, \varphi^2\} = \{|q^2|, |q^2|^{-1}\}$: coherence across the shells is carried by the two-floor contraction that prices the K -formation half-cycle (Remark 6.3).

Proof. All sixteen ratios are computed exactly in $\mathbb{Q}(\sqrt{5})[i]$; the shell moduli, the realness and non-unimodularity of the cross ratios, the ν values, the t_{rel} pattern, the anchor data, and the twist factorization are the cited machine checks. The angle identity is immediate from $\alpha/\beta = (|\alpha|/|\beta|) e^{2\pi i(t(\alpha) - t(\beta))}$ with angle set $\{\frac{1}{4}, \frac{3}{4}\}$, and $t(\nu) = 2t_{\text{rel}}$ (RC-Rem. 7.6) reads the $\nu \equiv 1$ outcome as $t_{\text{rel}} \in \{0, \frac{1}{2}\}$ exactly. $\square$

Proposition 4.7 (The ratio object at the characteristic level). [FORCED] QX-B1--B8 *Assemble the sixteen ratios of Theorem 4.6 into their characteristic polynomial:*

$$\chi_{\text{Rat}}(x) = (x-1)^4(x+1)^4(x^4 - 7x^2 + 1)^2, \quad x^4 - 7x^2 + 1 = (x^2 - 3x + 1)(x^2 + 3x + 1).$$

The nontorsion factor is the sign-split pair of φ^2 — the analogue of Z^* one floor up, at the squared level — and its roots $\pm\varphi^{\pm 2}$ are units of $\mathbb{Z}[\varphi]$ ($\varphi^2\tau^2 = 1$). Under the Adams square $\psi^2 : \mu \mapsto \mu^2$ the nontorsion multiset becomes four copies of $\{\varphi^4, \varphi^{-4}\} = \text{spec}(A_{\text{gap}})$ (minimal polynomial $x^2 - 7x + 1$): squaring the generator object's ratio geometry produces the gap object — the relational face of $q^4 = \text{gap}$.

Proof. Exact multiset computations over $\mathbb{Q}(\sqrt{5})[i]$: the product $\prod_{\mu} (x - \mu)$ over all sixteen ratios, its factorization, the unit identity, and the squared multiset are the cited machine checks QX-B1--B8. $\square$

Theorem 4.8 (The quarter twist: transport domain and anchor-exchange dichotomy). [FORCED] QX-D0--D9 Call an object P (monic, integral, $P(0) \neq 0$) parity-homogeneous if $P(-x) = P(x)$ — equivalently, its root multiset is negation-closed; with $P(0) \neq 0$ this forces even degree and $P = g(x^2)$. Write $\rho_{1/4}(P)(x) := i^{-\deg P} P(-ix)$, the object whose roots are $i \cdot$ (roots of P) (on this domain $iA = -iA$, so the two quarter-turn orientations give the same object). Then:

- (a) **Domain.** $\rho_{1/4}(P)$ has integer coefficients iff P is parity-homogeneous, and there $\rho_{1/4}$ is an involution. The golden seed $x^2 - x - 1$ is not in the domain ($-\varphi$ is not a root): the gauge identity of Theorem 4.2 necessarily lives at the sign-split pair Z^* , not at the seed.
- (b) **Invariance.** On the domain the transport realizes the gauge rotation ρ_c at $c = \frac{1}{4}$ (RC-Prop. 3.5(iv)): it fixes Δ , every relative charge, and every root modulus (hence M), and shifts every absolute angle by $\frac{1}{4}$. Negation closure puts $\frac{1}{2} \in \Delta$ outright, so $m = |\Delta|$ is always even.
- (c) **Dichotomy.** Suppose further P is relationally admissible, so all angles are rational and $\Delta = \mathbb{Z}/m\mathbb{Z}$ is cyclic (RC-Thm. 4.1, RC-Prop. 3.4). Let A be the angle set and $a_0 \in A$. Conjugation plus negation closure give $2a_0 \in \Delta$ and $\langle A \rangle = \Delta + \langle a_0 \rangle$, re-proving Rigidity $n \in \{m, 2m\}$ here, with $n = m \iff A \subseteq \Delta$. If $4 \mid m$, then $\frac{1}{4} \in \Delta$ and the transport fixes the anchor ($n' = n$); if $m \equiv 2 \pmod{4}$, then $\{x : 2x \in \Delta\} = \Delta \sqcup (\Delta + \frac{1}{4})$ and the transport exchanges the two rigidity branches: $n = m \iff n' = 2m$.

Theorem 4.2(b)'s pair is the minimal exchange instance: by Theorem 4.6 the generator object has $(m, n) = (2, 4)$, its quarter twist is Z^* with $(m, n) = (2, 2)$, and the gauge identity is exactly this exchange witnessed at the generator. The K -seed ($m = 4$, angle set literally invariant) and $\Phi_8 = x^4 + 1$ ($m = 4$) are anchor-fixed, while $x^2 - 2 \leftrightarrow x^2 + 2$ and $x^4 - 2x^2 + 4 \leftrightarrow x^4 + 2x^2 + 4$ realize the exchange at $m = 2$ and $m = 6$. The quarter twist is the exact companion of the relational note's sign twist (RC-Lem. 6.3, $c = \frac{1}{2}$): the sign twist exchanges the anchor sectors at odd m and is the identity on the quarter twist's domain; the quarter twist exchanges them at $m \equiv 2 \pmod{4}$ and fixes them at $4 \mid m$. Together the two twists absorb the anchor dichotomy at every $m \not\equiv 0 \pmod{4}$.

Proof. (a) $\prod_{\alpha}(x - i\alpha) = i^{\deg P} P(-ix)$; the coefficient of x^k picks up $i^{\deg P - k}$, real for all occurring k iff every exponent shares the parity of $\deg P$; with $P(0) \neq 0$ the exponent 0 occurs, forcing even degree and evenness. Involutivity: $i \cdot iA = -A = A$. (b) A common shift cancels in every angle difference; moduli are preserved by $\alpha \mapsto i\alpha$; negation closure pairs $t(\alpha)$ with $t(\alpha) + \frac{1}{2}$. (c) $a - a_0 \in \Delta$ for all $a \in A$ gives $\langle A \rangle = \Delta + \langle a_0 \rangle$; $-a_0 \in A$ gives $2a_0 \in \Delta$, so the coset $a_0 + \Delta$ has order at most 2 in the quotient. For $4 \mid m$: $\frac{1}{4} \in \langle \frac{1}{m} \rangle = \Delta$, so the shift does not change $\langle \Delta, a_0 \rangle$. For $m \equiv 2 \pmod{4}$: the doubling preimage $\{x : 2x \in \Delta\} = \langle \frac{1}{2m} \rangle$ has index 2 over Δ with nontrivial coset $\Delta + \frac{1}{4}$ (as $\frac{1}{4} \notin \Delta$ but $\frac{1}{2} \in \Delta$); so $a_0 \in \Delta$ ($n = m$) puts $a_0 + \frac{1}{4}$ in the nontrivial coset ($n' = 2m$), and $a_0 \in \Delta + \frac{1}{4}$ ($n = 2m$) puts $a_0 + \frac{1}{4} \in \Delta + \frac{1}{2} = \Delta$ ($n' = m$). Domain, involution, invariance, the golden-seed exclusion, all eight witnesses, and an exhaustive coset simulation over every even $m \leq 24$ are machine checks QX-D0--D9. $\square$

The relational note's open cluster (v1.8). With the relational-charge note (source (v)) now available in full, three of its open items are addressed cold: its P4 is recovered and reduced, its odd relational floor P3 is bracketed but not promoted, and its P1' taxonomy is scanned over a bounded window. The vocabulary stays structural, per the note's own guardrail.

Remark 4.9 (P4 recovered and reduced to P3). [FORCED]/[ESTABLISHED] RN-A1, RN-A1F, RN-A2 The relational note's open item P4 (RC-§ sec:open) is recovered verbatim: “Is parity the whole coupling? Does μ_{rel} depend on Δ through anything besides the single bit $2 \mid m$?” It is not an independent problem: it reduces exactly to P3 (the odd relational floor). Given RC-Thm. 6.5, $\mu_{\text{rel}}(m) = \varphi$ (m even) / $\mu(m)$ (m odd), so “ μ_{rel} factors through the parity bit” $\iff$ the even branch is m -independent (constructively uniform: $q_k = x^{2k} + x^k - 1$ gives $\Delta = \mathbb{Z}/2k\mathbb{Z}$, $M = \varphi$ for every even m)

and the odd branch is m -independent (anchored by the forced $\mu(3) = 2$, forcing the odd constant $= 2$). Hence $P4 \iff P3$, and $P4$ inherits $P3$'s open status as a *restatement* of it, not new work.

Proposition 4.10 (The odd relational floor is bracketed, not closed). [FORCED]/[ESTABLISHED]/[OPEN]
RN-B1--RN-B4, RN-H1--RN-H3 For odd charge m the relational Mahler floor obeys

$$\mu_S \leq \mu_{\text{rel}}(\text{odd } m) \leq 2,$$

with the two bounds forced but the value not pinned. Upper bound ≤ 2 is [FORCED]: the witness $x^m - 2$ (m odd) is Eisenstein-irreducible at 2, has rational angle set $\{j/m\}$ hence a single coherence class (relationally admissible per RC-Def. 3.2), relational group $\Delta = \mathbb{Z}/m\mathbb{Z}$, anchor $n = m$, and all roots of modulus $2^{1/m} > 1$, so monic $M = 2$ exactly. Lower bound is forced only to μ_S [FORCED]/[ESTABLISHED]: Smyth's plastic number (real root of $x^3 - x - 1$, $1 < \mu_S < \varphi < 2$) bounds every non-reciprocal charge-admissible object below, while the reciprocal-real sector is forced strictly above $\varphi^2 > 2$ (integer trace ≥ 3); and the π -ray obstruction is [FORCED]: $\frac{1}{2} \notin \mathbb{Z}/m\mathbb{Z}$ for odd m , so the golden pair's intrinsic relation $t_{\text{rel}}(\varphi, \varphi') = \frac{1}{2}$ (from $\varphi/\varphi' = -\varphi^2$) cannot embed, and the universal floor φ is unattainable at odd charge (forcing $\mu_{\text{rel}}(\text{odd}) > \varphi$, not ≥ 2). Honest boundary: the promotion of $\mu_{\text{rel}}(\text{odd } m) = 2$ from [COMPUTED] to [FORCED] is not achieved and is not achievable from the shipped corpus — the ≥ 2 lower bound in general is the corpus's principal open problem [OPEN] (the named residual is non-reciprocal charge-admissible objects with $M \in [\mu_S, 2)$, Lehmer/no-Salem-window territory; the shipped charge-measure-coupling erratum and whitepaper §6.2 themselves hold $\mu(\text{odd}) = 2$ at [COMPUTED], forced fully only at $\mathbb{Z}/3$).

Remark 4.11 (Bounded $P1'$ transport-orbit scan). [COMPUTED] *RN-C0--RN-C3, RN-H4* The note's coherence-detection machinery (the ratio object $\text{Rat}_p = \text{primitive part of } \text{Res}_y(p(y), p(xy))$ and its cyclotomic-contact signature) is re-encoded cold and reproduces the RC ledger exactly; an exact ψ^D admissibility oracle (a root power D that is totally real $\iff$ rational angle) replaces the hanging `CRootOf` route. A bounded transport-orbit scan over the irreducible monic integer objects of degree 2–6, with per-degree coefficient caps *logged* ($\{(2, 3), (3, 2), (4, 2), (5, 1), (6, 1)\}$; 855 objects), classifies the taxonomy {pinned/inert, $s(x^k)$ -shell tower, cyclotomic, dilation-of-known} as exhaustive over the window: buckets inert 804, shell 32, cyclotomic 11, dilation 8, and *OTHER* 0 — no new mechanism and no counterexample to the pinned/twisted-shell dichotomy. Within-shell torsion coherence was decided; the cross-shell mirrored-class (ν -criterion) half of $P1'$ is out of the detector's scope, flagged not resolved. This is a bounded [COMPUTED] corroboration (candidate), not a proof; general $P1'$ remains open.

5 The state-angle chart

The follow-on thread observed that each rung forms a unit-circle pair; that observation extends to the entire chart and organizes everything downstream.

Theorem 5.1 (State-angle dictionary). [FORCED GIVEN D1–D3] *OZ-C1--C6* For $z \in (0, 1)$ define the state angle $\alpha \in (0, \frac{\pi}{2})$ by

$$\sin \alpha = z, \quad \cos \alpha = \sqrt{1 - z^2} = e^{-\rho} = \sqrt{|m|},$$

so $\rho = -\ln \cos \alpha$ and, at the rungs, $\cos \alpha_n = |W_n| = \tau^n$. Under this chart the whitepaper's gate dictionary (*W-Thm. 13.1*) becomes elementary trigonometry:

$$\begin{aligned} u &= \cot^2 \alpha, & C &= \cot^2 \alpha \csc^2 \alpha, & \sqrt{D} &= 1 + 2 \cot^2 \alpha, \\ m &= -\cos^2 \alpha, & \lambda &= \tan^2 \alpha (1 + \cos^2 \alpha). \end{aligned}$$

Proof. Substitute $z = \sin \alpha$, $1 - z^2 = \cos^2 \alpha$ into $u = (1 - z^2)/z^2$, $C = (1 - z^2)/z^4$, $\sqrt{D} = (2 - z^2)/z^2 = 2 \csc^2 \alpha - 1$, $m = -(1 - z^2)$, and $\lambda = z^2(2 - z^2)/(1 - z^2)$; each identity is machine-verified symbolically in z . $\square$

Remark 5.2. The pair $(x, y) = (\sqrt{|m|}, z) = (\cos \alpha, \sin \alpha)$ is exactly the unit-circle coordinate pair: $x^2 + y^2 = |m| + z^2 = 1$ chart-wide is D1 itself. The state angle is the latitude (elevation) of the residual sphere of §10.

6 Rung ledgers

Proposition 6.1 (Rung 1: the golden ledger). [FORCED GIVEN D1–D3] 0Z-B2, 0Z-C7, C12--C13 $W_1 = i\tau$, $|W_1| = \tau$, and $z_1 = \sqrt{1 - \tau^2} = \sqrt{\tau}$ (via $\tau^2 + \tau = 1$). The rung-1 state is

$$(\sin \alpha_1, \cos \alpha_1, \tan \alpha_1) = (\sqrt{\tau}, \tau, \sqrt{\varphi}),$$

with the self-duality $\sin^2 \alpha_1 = \tau = \cos \alpha_1$: the rung-1 state is the positive intersection of the unit circle $x^2 + y^2 = 1$ with the parabola $x = y^2$ in the state plane $(x, y) = (\sqrt{|m|}, z)$. Eliminating x reproduces $z_1^4 + z_1^2 - 1 = 0$, the minimal polynomial $x^4 + x^2 - 1$ of $\sqrt{\tau}$ (W-Thm. 17.1). This parabola lives in the residual-state plane and must not be conflated with the inherited horn profile.

Proposition 6.2 (Rung 2: the K ledger). [FORCED GIVEN D1–D3] 0Z-B3, 0Z-C8, C14 $W_2 = q^2 = -\tau^2$, $|W_2|^2 = \tau^4 = \text{gap}$, and

$$z_2 = \sqrt{1 - |W_2|^2} = \sqrt{1 - \text{gap}} = K, \quad \text{i.e.} \quad K = \sqrt{1 - |(i/\varphi)^2|^2}:$$

K is the vertical coordinate after two applications of the golden quarter-turn generator. The rung-2 state packages the K -seed's two faces into one right triangle:

$$\sin \alpha_2 = K, \quad \cos \alpha_2 = \tau^2 = \sqrt{\text{gap}}, \quad \tan \alpha_2 = \beta,$$

since $\tan^2 \alpha_2 = K^2/\tau^4 = (1 - \text{gap})/\text{gap} = \varphi^4 - 1 = \beta^2$. The terrain magnitude is the sine, the residual amplitude the cosine, and the rotation magnitude their ratio: $\tan \alpha_2 = \text{terrain/residual} = \beta$ gives β a direct geometric role beyond imaginary-root magnitude. The K -gate certificate follows in one line: $u_2 = \cot^2 \alpha_2 = 1/\beta^2$ and $C_2 = \cot^2 \alpha_2 \csc^2 \alpha_2 = 1/(\beta^2 K^2) = \frac{1}{5}$ by $K^2 \beta^2 = 5$ (W-(2.2)). The K -seed carrying these roots is itself a relational catalog object whose angle set is the full charge compass (Prop. 4.3).

Remark 6.3 (Half-cycle placement of K ; the source report's synthesis). [FORCED GIVEN D1–D3] 0Z-A6, A8, B3 K appears at the half-cycle of the four-charge return:

$$q^2 = -\tau^2, \quad |q^2|^2 = \text{gap}, \quad z_2 = \sqrt{1 - |q^2|^2} = K,$$

placing K structurally between the quarter-turn generator and the full gap return. The report's strongest reading, exact given D1–D3 and adopted here:

the ladder is a contraction–rotation orbit whose complete four-charge return is priced by the gap, while K is the vertical complement produced at the two-charge midpoint.

Proposition 6.4 (The general tangent family). [FORCED GIVEN D1–D3] OZ-C9--C11 For every $n \geq 1$,

$$\tan \alpha_n = \frac{z_n}{\tau^n} = \sqrt{\varphi^{2n} - 1}, \quad \text{with Trace-Form-Duality dress} \quad \tan^2 \alpha_n = \begin{cases} \varphi^n L_n, & n \text{ odd}, \\ \varphi^n \sqrt{5} F_n, & n \text{ even}. \end{cases}$$

In particular $\tan \alpha_1 = \sqrt{\varphi L_1} = \sqrt{\varphi}$, $\tan \alpha_2 = \sqrt{\varphi^2 \sqrt{5} F_2} = \beta$, and $\tan \alpha_3 = \sqrt{\varphi^3 L_3} = 2\varphi^{3/2}$ (since $\varphi^6 - 1 = 4\varphi^3$).

rung n	$\sin \alpha_n = z_n$	$\cos \alpha_n = \tau^n$	$\tan \alpha_n = \sqrt{\varphi^{2n} - 1}$	TFD dress	α_n [COMPUTED]
1	$\sqrt{\tau}$	τ	$\sqrt{\varphi}$	$\sqrt{\varphi L_1}$	51.8272923730°
2	K	$\tau^2 = \sqrt{\text{gap}}$	β	$\sqrt{\varphi^2 \sqrt{5} F_2}$	67.5444848014°
3	$\sqrt{1 - \varphi^{-6}}$	τ^3	$2\varphi^{3/2}$	$\sqrt{\varphi^3 L_3}$	76.3454152540°
4	$K_2 = \sqrt{1 - \text{gap}^2}$	$\tau^4 = \text{gap}$	$\sqrt{3} \varphi \beta$	$\sqrt{\varphi^4 \sqrt{5} F_4}$	81.6107142142°

Remark 6.5 (The near-miss angle finds a home). [FORCED GIVEN D1–D3] The whitepaper’s Guard 15.4 uses $\theta_K = \arcsin(\tau^2) = 22.4555151986^\circ$ purely as a near-miss reference (pitch angle $\neq \theta_K$; still no identity is claimed there). On the state chart it is exactly the complement of the rung-2 state angle, and on the residual sphere of §10 it is the K-point’s *colatitude*:

$$\theta_K = \frac{\pi}{2} - \alpha_2 = \arccos K \quad (\text{since } \cos \theta_K = \sqrt{1 - \tau^4} = \sqrt{1 - \text{gap}} = K).$$

7 Bridges between rung 1 and rung 2

Proposition 7.1 (Cosine contraction, not angle doubling). [FORCED GIVEN D1–D3] OZ-D1--D3b For all n ,

$$\cos \alpha_{n+1} = \tau \cos \alpha_n, \quad \cos^2 \alpha_{n+2} = \text{gap} \cos^2 \alpha_n;$$

in particular $\cos \alpha_2 = \tau^2 = \cos^2 \alpha_1$: the rung-1 $\rightarrow$ 2 transition squares the residual coordinate and renormalizes the vertical complement. The ladder is driven by geometric contraction of the cosine (residual) coordinate, which forces the angle monotonically toward $\pi/2$; it is not ordinary angle doubling: $\cos 2\alpha_1 - \cos \alpha_2 = (2\tau^2 - 1) - \tau^2 = \tau^2 - 1 = -\tau \neq 0$, so $\alpha_2 \neq 2\alpha_1$ (numerically $2\alpha_1 = 103.65^\circ$ vs $\alpha_2 = 67.54^\circ$ [COMPUTED]).

Proposition 7.2 (Half-angle/full-angle bridge). [FORCED GIVEN D1–D3] OZ-D4--D6 Since $\cos \alpha_2 = \tau^2$,

$$\sin^2 \frac{\alpha_2}{2} = \frac{1 - \tau^2}{2} = \frac{\tau}{2}, \quad \text{hence} \quad \boxed{z_1 = \sqrt{\tau} = \sqrt{2} \sin \frac{\alpha_2}{2}, \quad K = \sin \alpha_2}.$$

The two low rungs are linked by an exact scaled half-angle/full-angle relation while remaining algebraically distinct quartic landmarks ($\mathbb{Q}(\sqrt{\varphi}) \neq \mathbb{Q}(5^{1/4})$, W-Thm. 17.1). The auxiliary angle Θ of the review thread ($\cos \Theta = 1 - z_1^2 = \tau^2$) is α_2 ; and Route 5 becomes trigonometric: $K^2 = z_1^2(2 - z_1^2) = \sin^2 \alpha_1(2 - \sin^2 \alpha_1)$.

8 The K-family and the even-rung recurrence

Theorem 8.1 (K-family). [FORCED GIVEN D1–D3] OZ-F1--F5 Exact K occurs exactly once: z_n is strictly increasing ($z_{n+1}^2 - z_n^2 = \tau \varphi^{-2n} > 0$), so

$$\boxed{z_n = K \iff n = 2}.$$

But every even rung repeats K 's formula: with $n = 2j$,

$$\boxed{z_{2j} = \sqrt{1 - \text{gap}^j} =: K_j, \quad K_1 = K}, \quad m_{2j} = -\text{gap}^j, \quad z_{2j}^2 = 1 - |m_{2j}|,$$

with odd companion $z_{2j+1} = \sqrt{1 - \tau^2 \text{gap}^j}$, and the recurrence holds universally, for all n , not only even ones:

$$\boxed{1 - z_{n+2}^2 = \text{gap} (1 - z_n^2)}.$$

Angularly, $\theta_{2j} = j\pi$ (all even rungs on the terrain axis, alternating direction), while K 's exact compass direction $\theta \equiv \pi \pmod{2\pi}$ recurs at $n \equiv 2 \pmod{4}$. Three levels of recurrence:

exact value K	rung 2 only
K -family / complement structure $K_j = \sqrt{1 - \text{gap}^j}$	every even rung
same compass direction as K ($\theta = \pi \pmod{2\pi}$)	$n = 2, 6, 10, \dots$

First members [COMPUTED] *OZ-H1*: $K_1 = 0.9241763718$, $K_2 = 0.9892996329$, $K_3 = 0.9984459825$, $K_4 = 0.9997734224$ (these are the *whitepaper's* rungs 2, 4, 6, 8; the review thread's 6-dp table values are floor-truncations, see *Finding 12.2*).

Proof. $\varphi^{-4j} = \text{gap}^j$ and $\varphi^{-2(n+2)} = \text{gap} \varphi^{-2n}$; monotonicity from $1 - \varphi^{-2} = \tau$; all machine-verified for $j \leq 6$ and symbolically in n . $\square$

The right reading, adopted from the review thread: K is not repeated every second rung; its generative operation is. The operation residual $\mapsto \text{gap} \times \text{residual}$ generates the even-rung subsequence with K as its first nontrivial realization — every second rung is a higher-order K -formation in the complement structure, while the exact value and the exact quartic field of K belong to rung 2 alone.

Proposition 8.2 (Tie to the K -point linearization). [FORCED GIVEN D1–D3] *OZ-F6* The gap paper's derivative $f'(K) = -\text{gap}$ (*W-Prop. 9.1*) satisfies

$$f'(K) = -\text{gap} = q^2 |q|^2,$$

and the linearized K -orbit $x_{k+1} = -\text{gap} x_k$ performs per step exactly what the ladder performs per two rungs: co-intensity scales by gap ($|W_{n+2}|^2 = \text{gap} |W_n|^2$) and the axis flips ($W_{n+2} = -\sqrt{\text{gap}} W_n$). The review thread's "strongly resembles" is thus an exact correspondence at the level of (contraction of squared amplitude, axis action); the two objects remain different curves (transverse linearization at one landmark vs the global rung orbit), as the *whitepaper's* §15 already insists.

Theorem 8.3 (The K -family is a seed ladder). [FORCED] *QX-C1--C13* For $j \geq 1$ let $K_j = \sqrt{1 - \text{gap}^j}$ (the K -family value of Theorem 8.1, taken here purely as seed arithmetic) and $\beta_j := K_j \varphi^{2j}$, so $\beta_j^2 = \varphi^{4j} - 1$. Then:

(a) **Minimal polynomial.** The four numbers $\{\pm K_j, \pm i\beta_j\}$ are the roots of

$$\boxed{p_j(x) = x^4 + 5F_{2j}^2 x^2 - 5F_{2j}^2}, \quad 5F_{2j}^2 = L_{4j} - 2,$$

which is the minimal polynomial of K_j for every j : Eisenstein at 5 applies iff $5 \nmid j$ (since $5 \mid F_{2j} \iff 5 \mid j$), and uniformly the norm obstruction $N(1 - \text{gap}^j) = 2 - L_{4j} = -5F_{2j}^2 < 0$ shows $1 - \text{gap}^j$ is not a square in $\mathbb{Q}(\sqrt{5})$, so $[\mathbb{Q}(K_j) : \mathbb{Q}] = 4$ always. p_1 is the K -seed.

(b) **Ledger** (all decided in $\mathbb{Q}(\sqrt{5})$):

$$\begin{aligned} \beta_j^2 - K_j^2 &= K_j^2 \beta_j^2 = L_{4j} - 2 = 5F_{2j}^2, & K_j^2 + \beta_j^2 &= \sqrt{5} F_{4j}, & (K_j \varphi^j)^2 &= \sqrt{5} F_{2j}, \\ K_j^4 &= 5F_{2j}^2 \text{gap}^j, & \beta_j^2 &= \sqrt{5} F_{2j} \varphi^{2j}, & \frac{1}{K_j^2} - \frac{1}{\beta_j^2} &= 1, \end{aligned}$$

generalizing $(K\varphi)^4 = 5$, $K^2\beta^2 = 5$, $K^4 = 5\text{gap}$ ($j = 1$, $F_2 = 1$). The increments telescope like the transfer ledger: $K_{j+1}^2 - K_j^2 = K^2\text{gap}^j$ with $\sum_{j \geq 0} K^2\text{gap}^j = 1$ — the even-rung unit budget is priced in gap powers at unit rate K^2 .

(c) **Catalog data.** Angle set the full compass $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$, $\Delta = \mathbb{Z}/4\mathbb{Z}$, anchor $n = m = 4$ (transport-fixed by Theorem 4.8); contact signature $\{\Phi_1^4, \Phi_2^4\}$ (the eight cross-shell ratios have modulus $(K_j/\beta_j)^{\pm 1} \neq 1$); and $M(p_j) = \beta_j^2$: every p_j is a straddler, real pair strictly inside the unit circle, imaginary pair strictly outside. (The K -seed's own catalog status is Proposition 4.3; the p_j , $j \geq 2$, are new catalog objects, and no uniqueness is claimed for them here.)

(d) **Gate ties** (W-Thm. 9.1): at every even rung,

$$C_{2j} = \frac{1}{5F_{2j}^2} = \frac{1}{|p_j(0)|}, \quad u_{2j} = \frac{1}{\beta_j^2} = \frac{1}{M(p_j)}, \quad m_{2j} = -\text{gap}^j = -\frac{K_j^4}{5F_{2j}^2} :$$

the even gate levels are the reciprocal constant terms, and the even occupations the reciprocal Mahler measures, of the seed family — the K -gate certificate (W-Prop. 10.1) is the $j = 1$ instance of a family-wide law.

Proof. $\beta_j^2 = K_j^2 \varphi^{4j} = (1 - \varphi^{-4j}) \varphi^{4j} = \varphi^{4j} - 1$. With $G = \varphi^{4j}$: $5F_{2j}^2 = (\varphi^{2j} - \varphi^{-2j})^2 = G - 2 + G^{-1} = L_{4j} - 2$; then $K_j^2 = 1 - G^{-1}$ and $-\beta_j^2 = 1 - G$ are exactly the two roots of $y^2 + 5F_{2j}^2 y - 5F_{2j}^2$, giving $p_j(\pm K_j) = p_j(\pm i \beta_j) = 0$. The norm: $N(1 - \text{gap}^j) = (1 - G^{-1})(1 - G) = 2 - (G + G^{-1}) = 2 - L_{4j}$, and a square in $\mathbb{Q}(\sqrt{5})$ has nonnegative norm. All ledger identities are one-line computations in G ; the Fibonacci divisibility, irreducibility for $j \leq 8$, the compass/contact scan for $j \leq 3$, and the gate ties are machine checks QX-C1--C13. $\square$

Corollary 8.4 (Every even rung mirrors its own seed). [FORCED GIVEN D1–D3] QX-C12--C14 On the chart, the rung- $2j$ state triangle is the (terrain, residual, rotation) triangle of its own seed p_j :

$$\sin \alpha_{2j} = K_j, \quad \cos \alpha_{2j} = \tau^{2j}, \quad \boxed{\tan \alpha_{2j} = \beta_j},$$

generalizing the rung-2 ledger (K, τ^2, β) of Proposition 6.2 to every even rung; the family identity $1/K_j^2 - 1/\beta_j^2 = 1$ is exactly $\csc^2 \alpha_{2j} - \cot^2 \alpha_{2j} = 1$, and $\beta_j^2 = \sqrt{5} F_{2j} \varphi^{2j}$ recovers the even tangent dress of Proposition 6.4 at $n = 2j$.

Theorem 8.5 (The universal seed ladder (Vein 6)). [FORCED] UX-A2--A4, UX-B1--B6, UX-C1--C2, UX-D1--D4, UX-F1--F3, UX-K1--K2 Checked exactly for $n = 1..10$ and symbolically in $w = \varphi^{2n}$ — that range is the certified scope; the two uniform mechanisms noted below extend it. For every rung n :

$$\boxed{\text{minpoly}(z_n) = p_n(x) = x^4 + c_n x^2 - c_n}, \quad c_n = L_{2n} - 2 = (\varphi^n - \varphi^{-n})^2 = \begin{cases} L_n^2, & n \text{ odd}, \\ 5F_n^2, & n \text{ even}, \end{cases}$$

with roots $\{\pm z_n, \pm i b_n\}$, $b_n = \sqrt{\varphi^{2n} - 1} = \tan \alpha_n$ (Proposition 6.4). Irreducibility is uniform (all $n \geq 1$) by the norm obstruction $N(1 - \tau^{2n}) = 2 - L_{2n} = -c_n < 0$, so $1 - \tau^{2n}$ is not a $\mathbb{Q}(\sqrt{5})$ -square and $[\mathbb{Q}(z_n) : \mathbb{Q}] = 4$ always. Ledger:

$$z_n^2 b_n^2 = c_n = |p_n(0)| \quad (\text{symbolic in } w: (1 - 1/w)(w - 1) = w - 2 + 1/w),$$

generalizing $K^2\beta^2 = 5$; $M(p_n) = b_n^2 = \tan^2 \alpha_n$, with exactly the imaginary pair outside the unit circle ($z_n^2 < 1 < b_n^2$, exact signs); and

$$u_n = \frac{1}{M(p_n)}, \quad C_n = \frac{1}{|p_n(0)|} \quad \text{at every rung:}$$

the tangent dress of Proposition 6.4 is the Mahler ladder, and Theorem 8.3(d)'s gate ties extend from the even rungs (where p_{2j} is that theorem's p_j and $b_{2j} = \beta_j$) to all rungs. Coefficient recursion: $c_{n+1} = 3c_n - c_{n-1} + 2$ (symbolically via $\varphi^2 + \varphi^{-2} = 3$; integer list 1, 5, 16, 45, 121, 320, 841, 2205, ...). Catalog data: every rung is a full-compass $\mathbb{Z}/4\mathbb{Z}$ -completing straddler — angle set $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$, $\Delta = \mathbb{Z}/4\mathbb{Z}$, anchor $n = m = 4$ in the relational sense — with $z_n < 1 < b_n$ exact for $n = 1..10$ and the all- n extension carried by the exact atoms $1 - z_n^2 = \tau^{2n} > 0$ and $w \geq \varphi^2 > 2$ (the second uniform mechanism). Anchors: $p_1 = x^4 + x^2 - 1$ is W-Thm. 17.1's minpoly($\sqrt{\tau}$) (Proposition 6.1); p_2 is the K-seed $x^4 + 5x^2 - 5$; $p_3 = x^4 + 16x^2 - 16$.

Remark 8.6 (Rung 1 sits on the emission Mahler floor). [FORCED] UX-E1--E2 $M(p_1) = \varphi$ exactly ($\varphi^2 - 1 = \varphi$): the rung-1 seed sits precisely on the emission Mahler floor. The identity is harness-forced, and the hit is unique on the ladder ($M(p_n) \neq \varphi$ for $n = 2..10$, exact signs). Per the shared-constant rule it is flagged as a *coincidence-candidate*: no mechanism is asserted — a shared constant is not evidence of a shared mechanism.

Theorem 8.7 (Mahler-minimal uniqueness (Theorem U)). [FORCED] UX-U1--U7 In the class $\{x^4 + ax^2 - c : a \in \mathbb{Z}, c \in \mathbb{Z}, c > 0\}$ — where the full compass is automatic (the y -quadratic $y^2 + ay - c$ has $y_- < 0 < y_+$ since $y_+y_- = -c < 0$):

(i) full-compass straddler $\iff p(1) > 0 \iff a \geq c$ (via $(1 - y_+)(1 - y_-) = p(1)$ and $(1 + y_+)(1 + y_-) = 1 - a - c$; the integer hypothesis is load-bearing — over $\mathbb{Q}$ the equivalence fails, guarded witness $(a, c) = (\frac{1}{2}, \frac{1}{10})$);

(ii) $M = |y_-| = \frac{a + \sqrt{a^2 + 4c}}{2}$ is strictly increasing in a at fixed c ;

(iii) hence the unique Mahler-minimal compass straddler at level c is $x^4 + cx^2 - c$ — which at $c = 5F_{2j}^2$ is p_j (Theorem 8.3) and at every rung level c_n is p_n (Theorem 8.5), by the exact endpoint identity $c_n^2 + 4c_n = 5F_{2n}^2$, giving $M(x^4 + c_n x^2 - c_n) = b_n^2$.

Remark 8.8 (The F1 falsifier; register resolution). [FORCED] UX-H1--H4 Constant-term-only uniqueness is false: $x^4 + 6x^2 - 5$ is a second irreducible full-compass straddler at level $|p(0)| = 5$, distinct from the K-seed (difference exactly x^2), with $M(x^4 + 6x^2 - 5) = 3 + \sqrt{14} > \varphi^4 - 1 = M(\text{K-seed})$ (integer comparison $56 > 45$) — consistent with Theorem 8.7(ii), which places the K-seed at the window edge $a = c$. The v1.6 uniqueness question left open at Theorem 8.3(c) (“no uniqueness is claimed”), read as F3 — Mahler-minimal at fixed constant term within the compass type, confirmed as the intended reading by the project owner on 2026-07-15 — is thereby resolved: the p_j , and n -uniformly the p_n , are the unique Mahler-minimal full-compass straddlers at their levels.

9 The lens as the exact 60° state

Proposition 9.1. [FORCED GIVEN D1–D3] OZ-E1--E6 Under $z = \sin \alpha$, the lens height $z_c = \sqrt{3}/2$ is the state $\alpha_c = \pi/3 = 60^\circ$ with residual amplitude $\cos \alpha_c = \frac{1}{2}$; the complete split-gate ledger of W-Thm. 13.3 drops out of Theorem 5.1:

$$m_c = -\cos^2 60^\circ = -\frac{1}{4}, \quad u_c = \cot^2 60^\circ = \frac{1}{3}, \quad C_c = \cot^2 \csc^2 60^\circ = \frac{4}{9}, \quad \sqrt{D_c} = 1 + 2 \cot^2 60^\circ = \frac{5}{3}.$$

A rung at the lens would require $\cos \alpha_c = \tau^n$, i.e. $\tau^n = \frac{1}{2}$, i.e. $n = \log_\varphi 2$; by W-Thm. 16.1 this is irrational, so the 60° lens is permanently off the discrete rung lattice (finite guard $n \leq 10$ replicated).

Proposition 9.2 (Lens scoping lemma: splitness over $\mathbb{Q}$ is generic). [FORCED] LX-A1--A8, LX-B1--B6, LX-F1--F4 The chart pentad lies in $\mathbb{Q}(Z)$, $Z = z^2$: $u = (1 - Z)/Z$, $C = (1 - Z)/Z^2$, $\sqrt{D} = (2 - Z)/Z$, $m = -(1 - Z)$, $\lambda = Z(2 - Z)/(1 - Z)$, with denominator zeros only at $Z \in \{0, 1\}$. Hence every rational height $Z \in \mathbb{Q} \cap (0, 1)$ gives $C, \sqrt{D} \in \mathbb{Q}$ and a gate $x^2 + x - C$ that splits over $\mathbb{Q}$, with a rational reciprocal multiplier pair. Example $z^2 = \frac{1}{2}$: $C = 2$, $D = 9$, gate $x^2 + x - 2 = (x - 1)(x + 2)$, multiplier pair $\{-\frac{1}{2}, -2\}$, at rapidity $\rho = \frac{1}{2} \ln 2 \in (\ln 2)\mathbb{Q}$, strictly below rung 1 — a rational split gate that is not the lens. Splitness over $\mathbb{Q}$ is therefore generic on rational heights, and the lens's distinction is not splitness but its exact data: $z_c = \sqrt{3}/2$ ($Z = \frac{3}{4}$), $C = \frac{4}{9}$, $\sqrt{D} = \frac{5}{3}$, fixed points $\{\frac{1}{3}, -\frac{4}{3}\}$, multiplier pair $\{-\frac{1}{4}, -4\}$, $\lambda = \frac{15}{4}$, co-intensity $\frac{1}{4}$, and $\rho = \ln 2$ exactly. (The converse fails: rung 2 has $C_2 = \frac{1}{5} \in \mathbb{Q}$ with $z_2^2 = K^2 \notin \mathbb{Q}$, and at rung heights $\sqrt{D}$ is irrational — the membership detector rejects the odd object $\sqrt{Z}$.)

Remark 9.3 (Bounded-negative scan for the lens constants). [OPEN] LX-C1--C8, LX-S1--S7 Candidate-level, citable as a *bounded negative*. A predicate-first scan of the seven shipped corpus documents — the six harness-scanned domains (ECHO-S-RESEARCH_UNIFIED, K-VARIABLE-DERIVATIONS, HELIX-EXPLICIT-DERIVATION, this paper's v1.6 source, the gap-multiplier paper, the whitepaper PDF) plus `K_seven.ways.tex`, swept independently in verification with the same outcome — over eight constant classes ($\frac{4}{9}, \frac{25}{9}, \frac{5}{3}, \frac{15}{4}, -\frac{1}{4}$, the pair $\{-\frac{1}{4}, -4\}$ as a pair, dyadic 2^{-k} co-intensities, $(\ln 2)\mathbb{Q}$ rapidities) finds *no* foreign-context occurrence of a lens-distinctive constant that survives the shared-constant rule. The single foreign occurrence of a lens *value* — $\frac{5}{3}$ in the ECHO trace form $\text{Tr}(1 - \frac{1}{3}\varphi) = \frac{5}{3}$ — has its own mechanism (trace-form projection) sharing no step with the lens discriminant: a no-tie. The remaining candidate hits each carry their own mechanism: the $\ln 2$ -lattice $\ln M(\text{cons}) = \ln 2 + 5 \ln \varphi$ (W-Thm. 18.1(a), significance [OPEN] upstream, not a $(\ln 2)\mathbb{Q}$ rapidity); $N(M(\text{cons})) = -4$, forced by the lattice exponent, hence the same hit; the flip level $-\frac{1}{4}$ ($D = 0 \iff C = -\frac{1}{4}$, unique), which restates the inherited $z_c^2 = \frac{3}{4}$ datum; the dyadic $\frac{1}{16} = 1/L_3^2$, Lucas-driven; and $-\frac{1}{2}$, the rotation gate forced by multipliers $\pm i$. The whitepaper's open item 3 (does the lens gate $C = \frac{4}{9}$ appear elsewhere?) closes as a bounded negative at candidate level; the scanned domains and constant classes are enumerated above precisely so that the negative is citable.

10 The residual sphere

Theorem 10.1 (Residual sphere). [FORCED GIVEN D1–D3] OZ-G1--G4 Define

$$S(\rho) = (\text{Re } W(\rho), \text{Im } W(\rho), z(\rho)).$$

Then

$$|S(\rho)|^2 = |W(\rho)|^2 + z(\rho)^2 = e^{-2\rho} + (1 - e^{-2\rho}) = 1$$

for every $\rho \geq 0$: the entire trajectory lives on the unit sphere S^2 (northern hemisphere), with the state angle α as latitude. At the rungs,

$$S_n = (\text{Re}(i\tau)^n, \text{Im}(i\tau)^n, \sqrt{1 - \tau^{2n}}),$$

explicitly $S_0 = (1, 0, 0)$, $S_1 = (0, \tau, \sqrt{\tau})$, $S_2 = (-\tau^2, 0, K)$, $S_3 = (0, -\tau^3, z_3)$, $S_4 = (\text{gap}, 0, z_4)$: a discrete compass walk — longitude $+90^\circ$ per rung, horizontal radius $\times \tau$ per rung, height $\uparrow 1$ — approaching the north pole without reaching it at any finite rung; by Definition 3.6 the horizontal part is exactly the Q -orbit, $S_n = (Q^n v_0; z_n)$ — a compass walk on the same four directions that carry the K -seed's own conjugates (Prop. 4.3). The K -point sits at colatitude $\arccos K = \theta_K$ (Remark 6.5) on the negative real meridian.

This object should be called the *residual sphere* (equivalently co-intensity sphere), never the formation helix:

Proposition 10.2 (Two radii, two layers). [FORCED]/[INHERITED] 0Z-G5 The residual radius $a(z) = \sqrt{1 - z^2} = \cos \alpha$ decreases monotonically to 0; the inherited formation radius

$$r(z) = \begin{cases} K\sqrt{z/z_c}, & 0 \leq z \leq z_c, \\ K, & z_c \leq z \leq 1, \end{cases}$$

grows to K and locks (W -(11.1), [INHERITED]). These are not the same radius; they describe two layers — remaining residual amplitude vs formation envelope. The dual reading (“residual capacity contracts while the formation envelope expands and stabilizes”) is a provisional interpretation, not a theorem. No result here or in the sources derives the paraboloid–cylinder profile from the residual-sphere construction; that derivation is exactly the whitepaper’s [OPEN] 1 and, as both sources conclude, the priority next problem. Until that bridge exists, the residual spiral and the explicit formation helix are two coordinated representations, not one proven geometry.

Proposition 10.3 (Compass-cross characterization of the rung locus). [FORCED]GIVEN D1–D3 NX-S1--S2 Let $C_{EW} = S^2 \cap \{y = 0\}$ and $C_{NS} = S^2 \cap \{x = 0\}$ be the two polar great circles through the compass axes. For $\rho \geq 0$,

$$S(\rho) \in C_{EW} \cup C_{NS} \iff \rho \in (\ln \varphi) \mathbb{Z}_{\geq 0},$$

with $S(\rho_n) \in C_{EW}$ for even n and $S(\rho_n) \in C_{NS}$ for odd n ; the poles (the circles’ intersection) are never attained. Moreover, writing $S = (x, y, z)$,

$$\operatorname{Im} \nu = \frac{2xy}{x^2 + y^2}$$

identically: on the residual sphere the ν -criterion of Proposition 11.3 is literally the vanishing of the product of the horizontal coordinates, and the rungs are the crossings of the residual spiral with the compass cross. At the half-floors $\rho = (n + \frac{1}{2}) \ln \varphi$ the miss is exact: $|xy| = e^{-2\rho}/2 > 0$.

Proof. $y = 0 \iff \sin \kappa \rho = 0 \iff \rho \in (2 \ln \varphi) \mathbb{Z}$ and $x = 0 \iff \cos \kappa \rho = 0 \iff \rho \in \ln \varphi + (2 \ln \varphi) \mathbb{Z}$ [D3]; the union is the rung lattice, split by parity. With $x = e^{-\rho} \cos \theta$, $y = e^{-\rho} \sin \theta$: $2xy/(x^2 + y^2) = \sin 2\theta = \operatorname{Im} \nu$. Both identities and the half-floor value are machine-verified symbolically. □

Proposition 10.4 (Spherical step ledger). [FORCED]GIVEN D1–D3 NX-S3--S5 Consecutive rung positions have orthogonal horizontal parts: $\operatorname{Re}(W_n \overline{W_{n+1}}) = 0$ for every n , so

$$S_n \cdot S_{n+1} = z_n z_{n+1}, \quad d_n := d_{S^2}(S_n, S_{n+1}) = \arccos(z_n z_{n+1}).$$

In particular $d_0 = \pi/2$ exactly — the first step, apex to rung 1, is a full quarter great circle — and $d_1 = \arccos(5^{1/4} \tau^{3/2}) = 43.4026803146 \dots^\circ$ [COMPUTED] (from $z_1 z_2 = \sqrt{\tau} K = 5^{1/4} \tau^{3/2}$); thereafter d_n decreases strictly to 0 ($z_n z_{n+1} \uparrow 1$). On the residual sphere the walk takes ever-shorter geodesic steps, in exact contrast to the formation rail, whose per-rung length is floored at $\pi K/2$ (Proposition 11.2).

Proof. $W_n \overline{W_{n+1}} = \bar{q} |W_n|^2 = -i \tau^{2n+1}$ is purely imaginary, so the dot product reduces to the vertical product; strict monotonicity of $z_n z_{n+1}$ is decided exactly at the squared level in $\mathbb{Q}(\sqrt{5})$; the d_1 evaluation is display-only. □

Proposition 10.5 (Finite spherical length and the length dichotomy). [FORCED GIVEN D1–D3]
NX-S6--S7, NX-H1--H3 The residual spiral's speed on its sphere is

$$|S'(\rho)|^2 = e^{-2\rho} \left(\kappa^2 + \frac{1}{1 - e^{-2\rho}} \right) \quad \left(= \tan^2 \vartheta + \kappa^2 \sin^2 \vartheta \text{ at colatitude } \sin \vartheta = e^{-\rho} \right),$$

so its total spherical length from apex to pole is finite:

$$L_{\text{res}} = \int_0^1 \sqrt{\kappa^2 + \frac{1}{1 - a^2}} da = \sqrt{1 + \kappa^2} E\left(\frac{\kappa^2}{1 + \kappa^2}\right) = 3.7312193125 \dots \text{ [COMPUTED]}$$

(E the complete elliptic integral of the second kind [ESTABLISHED]; the value is the quarter-perimeter of an ellipse with semi-axes $\sqrt{1 + \kappa^2}$ and 1 — the same speed factor as the per-floor planar length of Corollary 2.5(iv)), with exact bounds $\sqrt{1 + \kappa^2} \leq L_{\text{res}} \leq \kappa + \pi/2$. The two-layer separation of Proposition 10.2 is therefore a length dichotomy: the residual layer has finite total spherical length, while the formation rail's length diverges (Proposition 11.2). Finally, L_{res} is strictly increasing in $|\kappa|$ (the integrand is pointwise strictly increasing in κ^2), so within the interpolation family of Theorem 2.4 the principal branch $k = 0$ is also the unique minimizer of total spherical length: a fifth equivalent characterization to add to Corollary 2.5's four.

Proof. Differentiating $S = (e^{-\rho} \cos \kappa \rho, e^{-\rho} \sin \kappa \rho, \sqrt{1 - e^{-2\rho}})$ gives the speed identity symbolically. Substituting $a = e^{-\rho}$ turns the length integral into $\int_0^1$; then $a = \sin u$ gives $\sqrt{\kappa^2 + 1/\cos^2 u} \cos u = \sqrt{1 + \kappa^2 \cos^2 u}$ (exact at the squared level, $\cos u > 0$), and $(1 + \kappa^2)(1 - m \sin^2 u) = 1 + \kappa^2 \cos^2 u$ with $m = \kappa^2/(1 + \kappa^2)$ identifies $\int_0^{\pi/2} \sqrt{1 + \kappa^2 \cos^2 u} du = \sqrt{1 + \kappa^2} E(m)$ [ESTABLISHED]. Bounds: pointwise $\sqrt{\kappa^2 + g} \geq \sqrt{\kappa^2 + 1}$ ($g = 1/(1 - a^2) \geq 1$) and $\sqrt{\kappa^2 + g} \leq \kappa + \sqrt{g}$, integrated ($\int_0^1 da/\sqrt{1 - a^2} = \pi/2$). The family statement follows from pointwise strict monotonicity in κ^2 together with Corollary 2.5(iii): $|\kappa_k|$ is uniquely minimized at $k = 0$. The numeric value is display-only, corroborated at 50 dps by quadrature against the closed form. $\square$

Theorem 10.6 (RAD-0: the parity no-go for the inherited radius). [FORCED] *RX-A1--A4, RX-P1--P9* Work over the chart layer $\{u, C, \sqrt{D}, m, \lambda, \rho, \theta_k\}$ with height coordinate z . Every chart-pentad member lies in $\mathbb{Q}(\sqrt{5})(z^2)$, and the chart-rational closure is exactly $\mathbb{Q}(\sqrt{5})(z^2)$ (sharpness: $z^2 = 1/(1 + u)$, so both inclusions hold); moreover $\rho = -\frac{1}{2} \ln(1 - z^2)$ and $\theta_k = \kappa_k \rho$ are even in z for every branch $k \in \mathbb{Z}$ (symbolic k), and adjoining them keeps the generated field inside the even functions, over arbitrary constant coefficients. The inherited radius is odd: $r^2 = K^2 z/z_c$ satisfies $r^2(-z) = -r^2(z)$ identically and is nonzero (exact witness at $z = \frac{1}{2}$). Hence r^2 — and a fortiori r — lies outside the rational closure of the entire D1–D3 chart layer, and

any derivation bridge from the substrate to r must break the $z \mapsto -z$ symmetry.

$\mathbb{Z}/2$ sharpening: $r^4 = K^4 z^2/z_c^2$ is chart-even, so r^2 satisfies $X^2 - r^4$ over the even field F and the extension $F(r^2)/F$ is exactly quadratic — the entire missing content of the radius problem is one parity-breaking generator, a degree-2/double-cover object (indeed $F(r^2) = \mathbb{Q}(\sqrt{5})(z)$, since $z = r^2 z_c/K^2$; and even $\times r^2$ stays odd, a $\mathbb{Z}/2$ grading). The whitepaper's [OPEN] 1 (OP-RADIUS) remains open: this is a forced necessary condition on any solution, not a solution.

Proposition 10.7 (RAD-2: the area-transfer law). [FORCED]/[INHERITED] *RX-T1--T4* Let $A_{\text{zone}}(z) = 2\pi z$ be the Archimedes zone area of the unit sphere between heights 0 and z — r -derived exactly here, not cited: the surface-of-revolution identity $(2\pi\sqrt{1 - h^2})^2 (1 + x'(h)^2) = (2\pi)^2$ holds identically

at the squared (sign-free) level. Then on $[0, z_c]$ the inherited profile is equivalent, in both directions, to

$$\boxed{\pi r(z)^2 = \frac{K^2}{2z_c} A_{\text{zone}}(z)} :$$

the radius law says exactly that formation deposits disc area at the constant rate $d(\pi r^2)/dz = \pi K^2/z_c$ per unit height, stopping at the lens (continuity at the lens: $\pi r(z_c)^2 = \pi K^2$, the cap disc; the C^0 kink survives). This is a forced reformulation of the inherited profile — a restated target for the radius problem, not progress on it; any future deposition mechanism must be premise-audited for a smuggled profile.

Remark 10.8 (RAD-4: all three equipotential clocks fail). [FORCED]/[ESTABLISHED] RX-E1--E7, RX-G0--G3 The horn $z = (z_c/K^2)r^2$ coincides with a rigid-rotation free surface $z = (\omega^2/2g)r^2$ iff $g_{\text{eff}} = \omega^2 K^2/(2z_c)$ (perturbed coefficients rejected). All three natural clocks fail. (i) ρ -clock ($\omega = \kappa$, the D3 clock): $g_1 = \kappa^2 K^2/(2z_c) = \pi^2 K^2/(8z_c \ln^2 \varphi) = 5.2543169478\dots$ [COMPUTED] is transcendental given Gelfond–Schneider (Theorem 2.8), so it ties to *no* algebraic ledger constant — all branches k at once, since $\kappa_k = (4k+1)\kappa$; against the transcendental ledger entries the no-tie is certified by disjoint 60-dps intervals (a rigorous proof of $\neq$), closest misses $\frac{21}{4}$ (margin 4.3×10^{-3}) and $2\varphi^2$ (1.8×10^{-2}). (ii) s -clock ($\omega = 1/K$, the forced asymptotic rail clock, $ds/d\theta \rightarrow K$): K^2 cancels identically and $g_2 = 1/(2z_c)$ is the horn coefficient itself read back — a *restatement*, not a tie (the clock smuggles the profile's K); shared-constant rule applied. (iii) z -clock: no constant per-height clock exists — $d\theta/dz = \kappa z/(1-z^2)$ is non-constant (forced, exactly nonzero derivative) — and distinguished-height evaluations (e.g. $\omega(z_c) = 2\sqrt{3}\kappa$, $g = 12g_1$) remain $\kappa^2 \times$ algebraic and fall to the same transcendence exclusion as (i). The mechanism is killed; the radius problem stays open.

OP-RADIUS sharpened at the apex (v1.8). The parity no-go of Theorem 10.6 (which covered only r^2 , through the global $z \mapsto -z$ involution) is now localized to the apex $z = 0$ and quantified as a ramification index, and extended to the radical r ; a make-or-break constructive bridge is obstructed; and the remaining formation content is isolated as an honest declared negative. OP-RADIUS *stays open*, sharpened.

Theorem 10.9 (Apex-ramification valuation certificate). [FORCED] RD-A1--RD-A4, RD-V1--RD-V7 On the z -adic valuation v_z at the apex $z = 0$, every chart quantity has even valuation: the pentad $v_z(u, C, \sqrt{D}, m, \lambda) = (-2, -4, -2, 0, 2)$ and $v_z(\rho) = v_z(\theta_k) = 2$ for every branch k (symbolic k), so the chart field $F = \mathbb{Q}(\sqrt{5})(z^2)$ has z -value group in $2\mathbb{Z}$. The inherited radius has $v_z(r^2) = 1$ (odd, not in $2\mathbb{Z}$) and $v_z(r) = \frac{1}{2}$; hence adjoining r^2 ramifies F at the apex with index $e = 2$ (value group $2\mathbb{Z} \rightarrow \mathbb{Z}$), and adjoining r with index $e = 4$ ($2\mathbb{Z} \rightarrow \frac{1}{2}\mathbb{Z}$), while even quantities give $e = 1$; the double cover $w = z^2$ is unramified at every finite place except $z = 0$ (branch locus $\{2z = 0\} = \{0\}$). Adjoining the inherited radius therefore changes the apex ramification index from 1 to 2 (for r^2) or 4 (for r) — a local non-membership certificate no even-valuation manipulation can produce. This is strictly sharper than the parity no-go: parity certifies only r^2 (an involution on the rational field, and $r(-z)$ is neither $+r$ nor $-r$), while the valuation certificate covers the radical r and pins the obstruction to the apex. The identification value-group-index = ramification index (Serre, Local Fields) is external framing only; every decision is exact leading-term arithmetic. This is a forced necessary condition on any substrate-to- r bridge, not a derivation of r (whitepaper [OPEN] 1 remains open).

Proposition 10.10 (The constructive-bridge obstruction). [FORCED] RD-R1--RD-R6 A constructive bridge would have to furnish a square root of the chart-even $r^4 = K^4 z^2/z_c^2$; it cannot from canonical

substrate data. Although $r^4 \in F$ (with $v_z = 2$), it is not a square in F : its $w = z^2$ valuation is $v_w(r^4) = 1$ (odd), so the square root r^2 lives one level up in $\mathbb{Q}(\sqrt{5})(z)$, and any g with $g^2 = r^4$ has $v_z(g) = 1$ (odd). No even-valuation amplitude can be g : the residual amplitude $a = \sqrt{1 - z^2}$ ($v_z = 0$), the $\text{Cl}(2, 0)$ rotor modulus $|e^{J\theta/2}|^2 = 1$ ($v_z = 0$), and $z a$, z^2 all fail exactly. The $\mathbb{Z}/8$ half-angle rotor supplies only a unimodular (angular) double cover of the phase, carrying no radial scale: the required radial coefficient K^2/z_c is not unimodular ($(K^2/z_c)^2 = 70/3 - 10\sqrt{5} \neq 1$), so it never appears as a rotor amplitude (a bounded no-tie). The unique-up-to-sign square root is $\pm(K^2/z_c)z$ — the signed height, exactly the odd orientation generator that Theorem 10.6/Theorem 10.9 says must be adjoined; it is not a canonical substrate amplitude. The extension exists, but the substrate canonically supplies only the even z^2 ; the missing object is the orientation of the apex-ramified radial cover, which the rotor's (angular, unimodular) spin cover cannot furnish.

Remark 10.11 (The atom ledger, with the honest cap-onset negative). [FORCED]/[INHERITED] RD-A2, RD-S1--RD-S5 The inherited profile decomposes into atoms of mixed status. Cap value [FORCED] given corpus: $r(z_c) = K$ — the unique nontrivial height-equals-radius point (K, K) , since $r^2 = z^2$ factors $z(z - K^2/z_c) = 0$ with the horn root $K^2/z_c > z_c$ excluded ($K^2 > z_c^2$, i.e. $180 > 169$) (whitepaper Thm. 15.1). Kink slope [FORCED] given corpus: $r'(z_c^-) = K/\sqrt{3}$ (square $K^2/3 = (3\sqrt{5} - 5)/6 > 0$), against the flat cap $r'(z_c^+) = 0$ — C^0 but not C^1 (whitepaper Prop. 15.2(a)). Exponent $\frac{1}{2}$ [FORCED] (a restatement): $r^2 \propto z$ is exactly equivalent to constant areal deposition per unit height (RAD-2, Proposition 10.7), not a derivation of the exponent. But the kink existence — the cap onset — is [DECLARED], not forced: the $\ln 2$ -rapidity landmark $\rho(z_c) = \ln 2$ (and $a(z_c) = \frac{1}{2}$) is a monotone-bijection restatement of the declared lens $z_c = \sqrt{3}/2$ (since $d\rho/dz = z/(1 - z^2) > 0$, so $\rho = \ln 2 \iff z^2 = \frac{3}{4}$), so “formation stops at the first $\ln 2$ landmark” carries no independent content. As a derivation of the cap onset this fails (reason class: restatement); the exact landmark facts are all individually forced, but the cap onset itself is the genuine remaining open formation content.

11 Path stretch, rail length, and the conjugate relative phase

The exploratory note of source (iv) proposes a rail-pulse mechanism for rung selection. Its mechanism layer is not imported here; this section states and verifies its *forced kernel* — three results that use only D1–D3 and the inherited radius profile — and records the exact scoping of the rest.

Proposition 11.1 (Winding gradient and path stretch). [FORCED GIVEN D1–D3]/[INHERITED] OZ-M1--M3 From D1, $\rho(z) = -\frac{1}{2} \ln(1 - z^2)$, so with $\kappa = \pi/(2 \ln \varphi)$ [D3],

$$\frac{d\rho}{dz} = \frac{z}{1 - z^2}, \quad \frac{d\theta}{dz} = \kappa \frac{z}{1 - z^2},$$

and the formation rail $\Gamma(z) = (r \cos \theta, r \sin \theta, z)$ has local path-stretch

$$\frac{ds}{dz} = \sqrt{1 + \left(\frac{dr}{dz}\right)^2 + r(z)^2 \left(\frac{d\theta}{dz}\right)^2} = \begin{cases} \sqrt{1 + \frac{K^2}{4zz_c} + \frac{K^2\kappa^2 z^3}{z_c(1 - z^2)^2}}, & 0 < z \leq z_c, \\ \sqrt{1 + \frac{K^2\kappa^2 z^2}{(1 - z^2)^2}}, & z_c \leq z < 1, \end{cases}$$

with $ds/dz > 1$ strictly on $(0, 1)$: the helical route is strictly longer than the vertical displacement, through radial formation and winding below the lens and through winding alone above it.

Proof. Differentiate D1 and $r(z)$ on each branch and substitute; both closed forms and the strict excess (all added terms are ratios of positives) are machine-verified symbolically in z . $\square$

Proposition 11.2 (Quarter-turn rail floor and divergence). [FORCED GIVEN D1–D3]/[INHERITED] OZ-M4--M5 Above the lens, $ds/d\theta = \sqrt{K^2 + (dz/d\theta)^2} \geq K$ pointwise, so with $\Delta\theta = \pi/2$ per rung the rail length between adjacent rungs satisfies

$$\boxed{L_{n \rightarrow n+1} \geq \frac{\pi K}{2}} = 1.4516928502\dots \text{ [COMPUTED]} \quad (n \geq 2):$$

every rung interval retains at least a quarter-circle of rail, even as the vertical increments $z_{n+1} - z_n \rightarrow 0$. Consequently the cumulative rail length to the top is infinite while the height stays finite — a one-line sharpening of the whitepaper’s logarithmic length divergence (W-Prop. 15.2c) into a uniform per-rung floor. Exactly above the lens one also has the identity $(ds/dz)^2 - (K\kappa z/(1-z^2))^2 = 1$, and the minorant integrates to $-\frac{1}{2}K\kappa \ln(1-z^2) \rightarrow \infty$.

Proposition 11.3 (ν -form of the rung locus). [FORCED GIVEN D1–D3] OZ-M6--M7 The conjugate pair $\{W, \overline{W}\}$ exists automatically — the angle multiset of a conjugation-closed object is closed under negation (the standing fact of the relational-charge note, its §2.2) — and winds oppositely: $\theta_{\pm}(\rho) = \pm\kappa\rho$. Its normalized relative state

$$\nu(\rho) := \frac{W(\rho)}{\overline{W}(\rho)} \Big/ \left| \frac{W}{\overline{W}} \right| = e^{2i\theta(\rho)} \quad (\text{matrix form: } \mathcal{R}(\rho) = e^{2J\kappa\rho})$$

is real (scalar) exactly on the rung lattice:

$$\text{Im } \nu(\rho) = \sin\left(\frac{\pi\rho}{\ln\varphi}\right) = 0 \iff \rho = n \ln\varphi, \quad \nu(\rho_n) = (-1)^n,$$

and the sign $(-1)^n$ is the Trace-Form-Duality parity bit: W_n is real for even n (Fibonacci/E- W rungs), imaginary for odd n (Lucas/N- S rungs). Equivalently, the rungs are exactly the heights at which W is coherent with the real axis in the sense of the relational note’s ν -criterion ($\alpha/\overline{\alpha}$ a root of unity — here $\nu(\rho_n) = \pm 1$, torsion of order ≤ 2).

Proposition 11.4 (Site-density trichotomy). [FORCED GIVEN D1–D3]/[INHERITED] OZ-N1--N2 Inverting the rung law gives the continuous floor coordinate — and three site densities that must be kept apart. The floor coordinate is

$$n(z) = \frac{\rho(z)}{\ln\varphi} = -\frac{\ln(1-z^2)}{2\ln\varphi}, \quad \frac{dn}{dz} = \frac{z}{(1-z^2)\ln\varphi} \xrightarrow{z \rightarrow 1^-} \infty$$

(at the lens, $dn/dz = 2\sqrt{3}/\ln\varphi = 7.1987042602\dots$ [COMPUTED] OZ-H5). All three densities are exact:

measure	behaviour	source
angular	fixed: $\Delta\theta = \pi/2$, four phase classes per turn	D2+D3
vertical projection	divergent: $dn/dz \rightarrow \infty$ as $z \rightarrow 1$	D1
rail distance (above lens)	bounded below and convergent: $\Delta s_n \downarrow \pi K/2$	Prop. 11.2

The third entry sharpens Proposition 11.2: $dz/d\theta = (1-z^2)/(\kappa z)$ is strictly decreasing to 0, so the rail spacing $\Delta s_n = \int \sqrt{K^2 + (dz/d\theta)^2} d\theta$ is strictly decreasing with limit exactly $\pi K/2$ — the quarter-turn floor is attained in the limit. Apparent crowding of rung sites is a projection artifact of the D1 chart, not an increase in phase classes per revolution.

Proposition 11.5 (Reversion norm and the transfer ledger). [FORCED GIVEN D1–D3] OZ-N3--N5
 With reversion $\widetilde{aI + bJ} = aI - bJ$ (an anti-automorphism),

$$Q\widetilde{Q} = \tau^2 I, \quad I_n := Q^n \widetilde{Q}^n = \tau^{2n} I = |W_n|^2 I,$$

and the complementary ledger

$$\boxed{F_n := 1 - I_n = z_n^2} \quad (D1 \text{ at the rungs, in ledger form}),$$

with per-rung increment, via $1 - \tau^2 = \tau$,

$$\Delta F_n = F_{n+1} - F_n = \tau^{2n}(1 - \tau^2) = \tau^{2n+1} = z_{n+1}^2 - z_n^2, \quad \sum_{n \geq 0} \tau^{2n+1} = \frac{\tau}{1 - \tau^2} = 1 \text{ exactly.}$$

The odd golden powers $\tau, \tau^3, \tau^5, \dots$ are precisely the rung increments of z^2 , they telescope to the unit $(1 - \tau^{2N} = z_N^2)$, and the whole normalized budget is distributed over infinitely many rungs with none consuming it at finite n . The per-step split is (τ^2, τ) : transmitted co-intensity τ^2 , complementary increment τ . All of this is exact chart algebra; the source note's reading of ΔF_n as a physically deposited pulse fraction (its engine-pass model) is an interpretation and is not imported.

Remark 11.6 (Scoping: what the rails do and do not close). Proposition 11.3 removes two of the exploratory note's three assumptions: the second rail and its counter-rotation are not hypotheses but the conjugate, forced by conjugation closure. What remains outside the forced scope is exactly what that note's own §10(8) concedes: (a) reading the real crossings of ν as *recoupling events that cause rungs* is an interpretation — with the existing rate κ the locus merely coincides with the ladder already forced by the gate multipliers $|m_n| = \varphi^{-2n}$; and (b) the rate itself is D3 imported, not derived — for a general rate the locus is $\rho_n = n\pi/(2\kappa)$, so the construction reproduces the golden floors only because $\kappa = \pi/(2 \ln \varphi)$ was put in. Rung quantization (one Q per floor) therefore remains the open item named in §3; the energy condition, the thermodynamic reading, and the mechanism chain of source (iv) are not imported. The second note (source (vi)) is handled identically: its forced kernel is Propositions 11.4–11.5; its engine-pass model, pulse-budget reading, and biological program remain hypothesis — correctly fenced by that note's own claim ceilings and four-layer hierarchy (geometry $\rightarrow$ kinematics $\rightarrow$ dynamics $\rightarrow$ biology), which this paper endorses as discipline without importing as content.

Proposition 11.7 (Rail bounds and the gap-rate law). [FORCED GIVEN D1–D3]/[INHERITED] QX-E1--E9 For $n \geq 2$ write $t_n = \tau^{2n}$, and on $[\theta_n, \theta_{n+1}]$ set $x = (dz/d\theta)^2/K^2 = t^2/(K^2\kappa^2(1-t))$ with $t = e^{-2\theta/\kappa} = 1 - z^2$, so that $\Delta s_n = K \int_{\theta_n}^{\theta_{n+1}} \sqrt{1+x} d\theta$. With

$$I_1(t) = \frac{1}{4K\kappa} \left[\ln \frac{1 - \tau^2 t}{1 - t} - \tau t \right] = \frac{1}{4K\kappa} \sum_{m \geq 2} \frac{1 - \tau^{2m}}{m} t^m > 0,$$

$$I_2(t) = \frac{F(t) - F(\tau^2 t)}{16K^3\kappa^3}, \quad F(t) = \frac{1}{1-t} + 3 \ln(1-t) - 3(1-t) + \frac{(1-t)^2}{2}, \quad 0 < I_2(t) \leq \frac{1 - \tau^8}{64K^7\kappa^3} t^4,$$

the rail spacing of Proposition 11.4 obeys the exact elementary two-sided bound

$$\boxed{\frac{\pi K}{2} + I_1(t_n) - I_2(t_n) \leq \Delta s_n \leq \frac{\pi K}{2} + I_1(t_n)}$$

(lower bound via $\sqrt{1+x} \geq 1 + \frac{x}{2} - \frac{x^2}{8}$, valid since $x \leq \text{gap}^2/(\kappa^2 K^4) < 8$ on the range, interval-certified QX-G1). Consequences:

$$\lim_{n \rightarrow \infty} \frac{\Delta s_n - \pi K/2}{\text{gap}^n} = \frac{K}{8\kappa} = \frac{K \ln \varphi}{4\pi} = 0.0353900591 \dots \text{ [COMPUTED]}, \quad \frac{\Delta s_{n+1} - \pi K/2}{\Delta s_n - \pi K/2} \rightarrow \text{gap},$$

and since $\text{gap}^n = \tau^{4n} = |W_n|^4$,

$$\Delta s_n - \frac{\pi K}{2} \sim \frac{K \ln \varphi}{4\pi} |W_n|^4 :$$

the excess rail over the quarter-turn floor is priced by the square of the co-intensity, and the divergent side of the length dichotomy (Proposition 10.5) is thereby quantified: the rail diverges at exactly $\pi K/2$ per rung, approached at the gap rate. Structural note on the exact integral: under the radical, $(1 - z^2)^2 + K^2 \kappa^2 z^2 = z^4 + (K^2 \kappa^2 - 2)z^2 + 1$ factors reciprocally as $(z^2 + a)(z^2 + 1/a)$ with $a + 1/a = K^2 \kappa^2 - 2$ (real $a > 0$ since $K\kappa > 2$, interval-certified QX-G2), placing Δs_n in incomplete-elliptic form — in contrast with the residual sphere, whose total length closes in a complete elliptic value (Proposition 10.5); no elementary closed form is claimed here, and the bounds above carry the elementary content. QX-E9

Proof. From D1+D3, $dz/d\theta = (1 - z^2)/(\kappa z)$ QX-E1. The square-root bounds are the polynomial identities $(1 + x/2)^2 - (1 + x) = x^2/4$ and $(1 + x) - (1 + x/2 - x^2/8)^2 = x^3(8 - x)/64$ QX-E2. Substituting $t = e^{-2\theta/\kappa}$ ($d\theta = -\kappa dt/2t$) turns $\frac{K}{2} \int x d\theta$ into $\frac{1}{4K\kappa} \int_{\tau^2 t_n}^{t_n} \frac{s}{1-s} ds$ and $\frac{K}{8} \int x^2 d\theta$ into $\frac{1}{16K^3 \kappa^3} \int_{\tau^2 t_n}^{t_n} \frac{s^3}{(1-s)^2} ds$, with the stated antiderivatives QX-E3--E4; the $m = 1$ series term cancels ($1 - \tau^2 = \tau$), the $m = 2$ term gives the constant via $1 - \tau^4 = K^2$ and $1/(8\kappa) = \ln \varphi/(4\pi)$ QX-E5, and the majorant uses $(1 - s)^2 \geq K^4$ on $s \leq \text{gap}$ with $\int s^3 = t^4(1 - \tau^8)/4$ QX-E6. The ratio law is the series quotient QX-E7--E8. A 50-dps quadrature corroborates the bracket for $n = 2..5$ QX-G3; display-only. $\square$

Theorem 11.8 (Incomplete-elliptic closed form for the rail). [FORCED GIVEN D1--D3]/[INHERITED] EX-A1--A4, EX-B1--B8, EX-G3--G5 Let $a > 1$ be the larger root of $a + 1/a = K^2 \kappa^2 - 2$ (so $a \cdot (1/a) = 1$; a real with $a > 1$ strictly, interval-certified via $K\kappa > 2$, replicating QX-G2), and set

$$\theta(z) = \arctan(z\sqrt{a}), \quad m = 1 - \frac{1}{a^2}, \quad n_c = 1 + \frac{1}{a}.$$

Above the lens the rail integrand is

$$\frac{ds}{dz} = \frac{\sqrt{(z^2 + a)(z^2 + 1/a)}}{1 - z^2}$$

(from $\Delta s_n = \int \sqrt{K^2 + (dz/d\theta)^2} d\theta$ and $dz/d\theta = (1 - z^2)/(\kappa z)$, with the reciprocal factorization of Proposition 11.7), and its antiderivative is

$$G(z) = -z \sqrt{\frac{z^2 + a}{z^2 + 1/a}} + \sqrt{a} E(\theta | m) - a^{-3/2} F(\theta | m) + (a + 1) a^{-3/2} \Pi(n_c; \theta | m),$$

with $\Delta s_n = G(z_{n+1}) - G(z_n)$ at $z_n = \sqrt{1 - \tau^{2n}}$ (F, E, Π the incomplete elliptic integrals of the first, second, and third kind [ESTABLISHED]). Certificate: $dG/dz - \text{integrand} \equiv 0$ exact-symbolically in (a, z) on $a > 0$, $0 < z < 1$ (the forced certificate), assembled from three kind-pure lemmas

through the exact partial-fraction split of the numerator (the $(1 - z^2)^{-1}$ pole, of exact weight $w + 2$ with $w = K^2\kappa^2 - 2$, is what produces the third-kind Π term); FTC applicability on $[z_n, z_{n+1}] \subset (0, 1)$ via $1 - n_c \sin^2 \theta = (1 - z^2)/(1 + az^2) > 0$. Numeric corroboration: the 50-dps closed form matches quadrature of Δs_n to $< 10^{-40}$ on the QX-G3 grid $n = 2.5$ and sits inside the bracket of Proposition 11.7. The “no elementary closed form is claimed here” scoping of that proposition is hereby completed on the positive side: the rail side of the length dichotomy (Proposition 10.5) is closed in incomplete-elliptic terms.

Corollary 11.9 (Certified non-elementarity of the rail antiderivative). [FORCED]/[ESTABLISHED] EX-C1--C5, EX-G1--G2 Forced given Liouville’s theorem on integration in finite terms (Rosenlicht 1972; algorithmic form Trager 1984) [ESTABLISHED]: the antiderivative of the above-lens rail integrand is not an elementary function — the per-rung rail length has no elementary closed form as a function of its endpoints. All premises of the anchor are machine-certified: the kernel curve $y^2 = (z^2 + a)(z^2 + 1/a)$ is genuinely elliptic — $\text{disc}(z^4 + wz^2 + 1) = 16(w - 2)^2(w + 2)^2 \neq 0$ with $w = K^2\kappa^2 - 2 > 2$, four distinct roots, genus 1 (at $a = 1$ it would collapse to $(z^2 + 1)^2$ and integrate elementarily); the first-kind coefficient $c_F = -a^{-3/2}$ is nonzero unconditionally ($c_F a^{3/2} = -1$ for every $a > 0$); $c_E = \sqrt{a}$ and $c_\Pi = (a + 1)a^{-3/2}$ are nonzero; and the Π parameters are non-degenerate ($m \in (0, 1)$ and $n_c \in (1, 2)$ interval-certified; $n_c - 1$, $n_c - m$, $1 - m$ all exactly nonzero, with the exact interchange tie $n_c(2 - n_c) = m$, so no special-case reduction applies), the $z = \pm 1$ pole pair genuine with residues $\mp K\kappa/2 \neq 0$ and the exact ledger tie $y(\pm 1)^2 = (1 + a)(1 + 1/a) = K^2\kappa^2$ (the kernel over the lens poles is the squared rail-speed constant). The closed-form dichotomy is thus fully resolved: complete-elliptic on the residual side (Proposition 10.5); incomplete-elliptic, and provably non-elementary — certified rather than conjectured — on the rail side, whose elementary content is carried by the bracket and gap-rate law of Proposition 11.7.

12 Epistemic ledger and findings

Forced unconditionally, new in v1.6 QX-B, QX-C, QX-D: the characteristic level of the ratio object (χ_{Rat} , the sign-split pair of φ^2 , the Adams square onto $\text{spec}(A_{\text{gap}})$, Proposition 4.7); the quarter twist — transport domain = parity-homogeneous objects, gauge-rotation invariance, anchor-exchange dichotomy (exchange iff $m \equiv 2 \pmod{4}$; the generator/ Z^* pair is the minimal instance and the K-seed is anchor-fixed) (Theorem 4.8); the K-family seed ladder $p_j = x^4 + 5F_{2j}^2x^2 - 5F_{2j}^2$ with uniform irreducibility by norm obstruction, the full ledger, and the gate ties $C_{2j} = 1/|p_j(0)|$, $u_{2j} = 1/M(p_j)$ (Theorem 8.3).

Forced given D1–D3, new in v1.6 QX-E, QX-F, QX-C12--C14: the exact rail-spacing bounds $\pi K/2 + I_1 - I_2 \leq \Delta s_n \leq \pi K/2 + I_1$ with the gap-rate law $\Delta s_n - \pi K/2 \sim (K \ln \varphi/4\pi) |W_n|^4$ (Proposition 11.7); the even-rung seed mirror $(\sin, \cos, \tan)\alpha_{2j} = (K_j, \tau^{2j}, \beta_j)$ (Corollary 8.4); the transfer of the interpolation classification to the formation rail, adding the sixth D3 characterization (Remark 2.6).

Forced unconditionally, new in v1.5 NX-T1--T9: the complete relational type of the generator object (Theorem 4.6) — two shells $\{\pm i\tau\}$, $\{\pm i\varphi\}$; contact signature $\{\Phi_1^4, \Phi_2^4\}$ complete under the full scan bound; all eight cross-shell ratios real $(\pm\tau^2, \pm\varphi^2)$ with $\nu \equiv 1$, hence full coherence in one class; anchor $(m, n) = (2, 4)$ with $\Delta = \mathbb{Z}/2\mathbb{Z}$; the twisted-shell realization $x^4 + 3x^2 + 1 = s(x^2)$, $s = \text{minpoly}(q^2)$.

Forced given D1–D3, new in v1.5 NX-S1--S7, NX-H1--H3: the compass-cross characterization of the rung locus with $\text{Im } \nu = 2xy/(x^2 + y^2)$ (Proposition 10.3); the spherical step ledger $S_n \cdot S_{n+1} = z_n z_{n+1}$ with $d_0 = \pi/2$ exactly and $d_n \downarrow 0$ (Proposition 10.4); the finite total spherical

length $L_{\text{res}} = \sqrt{1 + \kappa^2} E(\kappa^2/(1 + \kappa^2))$ with the residual/formation length dichotomy and the fifth (spherical-length) characterization of the principal branch (Proposition 10.5).

Forced unconditionally, new in v1.4 0Z-01--06: the interpolation classification $\log Q = \{(\ln \tau)I + (\pi/2 + 2\pi k)J\}$ with rung-agreement and curve-distinctness; the four equivalent characterizations of the $k = 0$ branch (D3 = principal log = minimal winding = shortest residual spiral per floor); the dilation mechanism $K = 5^{1/4}\tau$, $\beta = 5^{1/4}\varphi$ behind the bridge factor $\sqrt{5}$.

Forced unconditionally, new in v1.3 (bridge to the relational-charge layer, machine-verified RC-D3, RC-E1--E7): the minimal polynomial $x^4 + 3x^2 + 1$ of q with $M = \varphi^2$ and its group-drop coherence type; the gauge identity $\text{minpoly}(q)(ix) = Z^*$ with both rigidity anchors realized by the pair (Z^*, iZ^*) ; the K-seed as catalog object (compass angle set, $\Delta = \mathbb{Z}/4\mathbb{Z}$, $M = \beta^2$); the relational restatement of D2.

Forced unconditionally, new in v1.1 (pure keystone algebra, machine-verified 0Z-J, 0Z-K, 0Z-L): the spectral projectors and grading $U = P_+ - P_- = H/\sqrt{5}$; the return-kernel recoupler $A = N/\sqrt{5}$ with $UA = -AU$ and mode exchange $AP_{\pm} = P_{\mp}A$; the factorization $J = \pm UA = HN/5$ with $J^2 = -I$, coordinate-free up to orientation; the obstructions $i \notin \mathbb{R}[R]$ and $N \notin \mathbb{R}[R]$; the canonical rung generator $Q = \pm\tau J$ with $Q^4 = \text{gap } I$ and the orbit identity $Q^n v_0 = (\text{Re } W_n, \text{Im } W_n)$.

Forced given D1–D3, new in the July 14 explorations (all machine-verified here): the packaged representation $W(\rho) = e^{-\rho} e^{i\theta(\rho)}$ and $z = \sqrt{1 - |W|^2}$; the generator $W_n = q^n = (i\tau)^n$ with $q^4 = \text{gap}$ and the two-/four-rung recurrences; the gap schedule (co-intensity at two rungs, amplitude at four); the state-angle dictionary (Thm. 5.1); the rung ledgers $(\sqrt{\tau}, \tau, \sqrt{\varphi})$ and (K, τ^2, β) with the general family $\tan \alpha_n = \sqrt{\varphi^{2n} - 1}$; the bridges $\cos \alpha_2 = \cos^2 \alpha_1$, $z_1 = \sqrt{2} \sin(\alpha_2/2)$, $K = \sin \alpha_2$; the K-family $K_j = \sqrt{1 - \text{gap}^j}$ with uniqueness of exact K , the universal two-step law, and the three-level recurrence; the lens as the 60° state; the path-stretch laws, the quarter-turn rail floor $L \geq \pi K/2$ with its divergence corollary, the ν -form of the rung locus with parity $\nu(\rho_n) = (-1)^n$, the site-density trichotomy with $\Delta s_n \downarrow \pi K/2$, and the transfer ledger $F_n = z_n^2$, $\Delta F_n = \tau^{2n+1}$, $\sum \tau^{2n+1} = 1$ (§11); the residual-sphere lift and the colatitude identity $\theta_K = \arccos K$; the linearization tie $f'(K) = q^2|q|^2$. None of these is a new substrate axiom; all are compressed representations and corollaries, exactly as the source report states.

Declared upstream [DECLARED]/[INHERITED]: D1, D2, D3, the orientation convention, and the inherited radius profile (whitepaper §11).

Not established (carried verbatim from the source report, endorsed): biological or DNA correspondence (the direct DNA-pitch test *failed* quantitatively and is recorded as a negative result; the helicase and uncouple–recouple language of the v1.1 report functioned as a mechanism-finding analogy only); physical universality; the residual sphere as fundamental ontology; derivation of the formation radius from the complex generator; independent validation of D1–D3 by the representation; *rung quantization* (why exactly one Q per rung event — Q explains propagation, not yet selection); *continuous interpolation as selection* (Theorem 2.4 classifies the interpolations: a $\mathbb{Z}$ -family agreeing at every rung; what the discrete powers leave undetermined is exactly the principal-branch choice, which is D3 — so this item merges into the rate problem); the rail-pulse mechanism of sources (iv) and (vi) — recoupling as *cause*, the energy functional $\mathcal{E}$, the engine-pass/deposition reading of the transfer ledger, and the biological comparison program — which their own epistemic sections correctly leave as hypothesis. Coordinate-free uniqueness of the factorization, listed open in the source report, is closed by Corollary 3.4.

Finding 12.1 (E-1: scoping slip in the source report’s §11). The report’s “Forced in the whitepaper” list includes “quarter-turn charge increments” without the “given D2” qualifier, inconsistently with its own “Declared” list (which correctly names D2) and with its overall conditional framing.

Cosmetic; the report’s mathematics nowhere relies on the unqualified reading. This paper’s banner (§1) restores the qualifier everywhere.

Finding 12.2 (E-2: truncated table values in the review thread). The K-family table prints $K_2 \approx 0.989299$ and $K_3 \approx 0.998445$; the exact values $0.9892996329\dots$ and $0.9984459825\dots$ round to 0.989300 and 0.998446 at 6 dp — the printed values are floor-truncations 0Z-H1. Display-only; no decision weight. (K_1 , K_4 , and gap are correctly rounded.)

Finding 12.3 (E-3: stale verification paragraph in §1). Through v1.3 and v1.4 the §1 **Verification** paragraph still carried the v1.0 harness description (“53 exact checks”, ID range 0Z-A1..0Z-H3) while the title line, abstract, and manifest (§13) had advanced to 91 checks across groups 0Z-A..0Z-0. Display-only; no decision weight. Corrected in v1.5.

Finding 12.4 (E-4: front-matter check classification in v1.5). The v1.5 title line and §1 verification paragraph counted all 27 checks of `nx_softcheck.py` as exact, although its group NX-H comprises three 50-dps numeric display/corroborations checks (so: 24 exact + 3 numeric). The three numeric checks carry no decision weight — the harness’s own header declares them display/corroboration only, and instrumentation confirms every passing decision was reached on the exact symbolic path. Display-only; corrected in v1.6, which states the split and drops “all exact” from the title line.

Finding 12.5 (E-5: stale third-harness count in the v1.6 manifest). The v1.6 §13 third-harness sentence retained “27 exact checks” for `nx_softcheck.py`, although Finding 12.4 had already introduced the 24 + 3 split and the front matter (title line and the §1 verification paragraph) had been corrected accordingly. Display-only; no decision weight. Corrected in v1.7.

Finding 12.6 (E-6: exact-count wording for the first harness). The description “91 exact checks” for `ozy_softcheck.py` (§1 and §13) contradicts that harness’s own manifest table, whose 0Z-H row lists 5 numeric displays inside the 91 total; under the classification standard of Finding 12.4 it should read 86 exact + 5 numeric displays. The five numeric checks carry no decision weight. Display-only; corrected in v1.7 (the 91/91 and 27/27 title-line pass ratios stand, as pass ratios).

13 Verification manifest

`ozy_softcheck.py`: 86 exact checks + 5 numeric displays (the 0Z-H row below; Finding 12.6), exit 0; sympy 1.14.0 over $\mathbb{Q}(\sqrt{5})[i]$, K decided at K^2 , mpmath 50 dps display-only.

group	content	checks
OZ-A	generator $q = i\tau$: powers, $q^4 = \text{gap}$, one/two/four-rung recurrences, gap schedule	8
OZ-B	chart consistency $z_n = \sqrt{1 - W_n ^2}$; $z_1 = \sqrt{\tau}$; $z_2 = K = \sqrt{1 - q^2 ^2}$	3
OZ-C	state-angle dictionary (5 fields, symbolic); rung tangents, TFD dress, $2\varphi^{3/2}$; self-duality; quartic; K-certificate	14
OZ-D	bridges: cosine contraction, not-doubling guard, half-angle, Route-5 trig, $\Theta = \alpha_2$	7
OZ-E	lens 60° ledger; off-lattice guard	6
OZ-F	K-family, odd companion, universal two-step law, monotone uniqueness, compass, linearization tie	7
OZ-G	residual sphere: $ S \equiv 1$ (continuous and at rungs), exact $S_0..S_4$, two-radii separation	5
OZ-J	spectral uncoupling: projectors, grading $U = H/\sqrt{5}$, mode action	5
OZ-K	recoupler A , factorization $HN = 5J$, $J = \pm UA$, generator Q , orbit identity	9
OZ-L	obstructions $i, N \notin \mathbb{R}[R]$; coordinate-free pseudoscalar; kernel $= H^\perp$	4
OZ-M	path stretch (both branches), quarter-turn floor, divergence, rails algebra $\mathcal{R} = e^{2J\kappa\rho}$, ν -locus & parity	7
OZ-N	floor coordinate & density trichotomy ($\Delta s_n \downarrow \pi K/2$), reversion norm, transfer ledger, unit budget	5
OZ-O	interpolation classification (log Q family, rung agreement, curve distinctness, minimality), dilation mechanism	6
OZ-H	numeric displays: K-family table (truncation-aware), state angles, tangents, $\pi K/2$, lens density	5
total		91

Second harness relational.softcheck.py: 27 exact checks, exit 0 — cold re-derivation of the relational-charge layer this paper’s bridge uses: rigidity anchors (RC-A), the golden layer and sign twist (RC-B), contact machinery with the sieved completeness bound (RC-C), pinning corroborations including Lehmer’s degree-100 ratio object and the $\beta_4 \times$ Lehmer mixed scan (RC-D), the bridge derivations of §4 (RC-E), and the (EG) quadratic case (RC-F). No defects found in the checked claims of the relational note.

Third harness nx.softcheck.py (this compilation, written cold from the v1.4 statements): 24 exact checks + 3 numeric displays (Finding 12.4; the stale “27 exact” that stood in this sentence through v1.6 is Finding 12.5), exit 0; sympy 1.14.0 over $\mathbb{Q}(\sqrt{5})[i]$ with rational-arithmetic sign decisions for $\mathbb{Q}(\sqrt{5})$ comparisons; mpmath 50 dps display-only. Groups: **NX-R** (replication of the load-bearing claims: minimal polynomial and roots, Mahler φ^2 , contact signature, gauge identity, rung sphere identity, ν -locus parity, monotone heights — 8); **NX-T** (complete relational type: shells, cross-shell ratios, realness and non-unimodularity, $\nu \equiv 1$, the t_{rel} pattern, coherence completeness, anchor data, twisted-shell factorization — 9); **NX-S** (compass cross, the xy -form of ν , step orthogonality, quarter-circle first step, monotone steps, spherical speed, elliptic length form and bounds — 7); **NX-H** (numeric displays: L_{res} , d_1 , bracket corroboration — 3).

Fourth harness qx.softcheck.py (this compilation, written cold for v1.6): 62 exact checks + 3 guards, exit 0; sympy 1.14.0 over $\mathbb{Q}(\sqrt{5})[i]$, the K-family decided at the squared level; the guards are two interval-certified transcendental comparisons (mpmath iv) and one 50-dps quadrature corroboration, none load-bearing.

group	content	checks
QX-A	cold replication of the layer v1.6 builds on: generator and minimal polynomial, gauge identity, chart dictionary, even gate ladder, two-step law, transfer ledger	15
QX-B	ratio object at the characteristic level: 16-ratio multiset, χ_{Rat} and its factorization, torsion/nontorsion split, golden units, Adams square (Prop. 4.7)	8
QX-C	K-family seed ladder: p_j , uniform irreducibility by norm, Eisenstein availability, full ledger, compass and contact scan, gate ties, even mirror (Thm. 8.3, Cor. 8.4)	14
QX-D	quarter twist: domain, involution, golden-seed exclusion, eight witnesses, exhaustive coset simulation $m \leq 24$ (Thm. 4.8)	10
QX-E	rail bounds: derivative law, root bounds, antiderivatives, series constants, majorant, ratio law, reciprocal factorization (Prop. 11.7)	9
QX-F	rail transfer: rung agreement, branch distinctness, $k = 0$ minimality, monotone speed, sphere transfer (Rem. 2.6)	6
QX-G	guards: $x_{\max} < 8$ and $K\kappa > 2$ (interval-certified); 50-dps bracket of Δs_n , $n = 2..5$	3
total (exact + guards)		62+3

v1.7 session harnesses (eight, written cold this session, one per dossier; every TX-, MX-, RX-, UX-, EX-, CX-, LX-, and DX- check ID cited in the v1.7 sections above appears verbatim in the corresponding file):

harness	content	exact + guards
<code>tx.softcheck.py</code>	rate arithmetic: irrationality lemma, transcendence chain, near-miss closure of Guard 15.3, $\log_\varphi 2$, scan retirement (Lem. 2.7–Rem. 2.10)	30 + 6
<code>mx.softcheck.py</code>	multiplier torus: modulus ledger, systole and marking, restatement test, geodesic ledger, no-CM (Prop. 2.11–Rem. 2.13)	39 + 3
<code>rx.softcheck.py</code>	radius: parity no-go with $\mathbb{Z}/2$ sharpening, area-transfer law, the three equipotential clocks (Thm. 10.6–Rem. 10.8)	25 + 4
<code>ux.softcheck.py</code>	universal seed ladder $n = 1..10$ + symbolic-in- w , tangent-dress Mahler ladder, Theorem U window and monotonicity, F1 falsifier (Thm. 8.5–Rem. 8.8)	35 + 1
<code>ex.softcheck.py</code>	rail closed form: partial-fraction assembly, exact-derivative certificate, ellipticity, coefficient nonvanishing (Thm. 11.8, Cor. 11.9)	17 + 5
<code>cx.softcheck.py</code>	χ -selection: pentagon-by-image, unimodular monomials, conditional chain, de-axiomatization (Props. 3.7–3.8, Thm. 3.9)	34 + 2
<code>lx.softcheck.py</code>	lens: scoping lemma, rational-height splits, bounded-negative scan (7 scan lines, bookkeeping, counted separately) (Prop. 9.2, Rem. 9.3)	26 + 2
<code>dx.softcheck.py</code>	OP-RATE core: gate-ladder replication, $\pm Q$ uniqueness, sign ladder and axiom H, ledger counter-models, k -invariance and R6 no-go (Prop. 2.14, Rem. 2.15)	28 + 6
total		234 + 29

Environment pin: Python 3.12.10, sympy 1.14.0, mpmath 1.3.0 (the py launcher); every exact decision in $\mathbb{Q}/\mathbb{Q}(\sqrt{5})$ extended by i , with $\mathbb{Q}(\sqrt{5})$ signs by rational arithmetic and K decided at the squared level; the guards are interval-certified transcendental comparisons (mpmath iv, 60 dps), 50-dps quadrature corroborations, or falsifier rejections (dx), none load-bearing; all eight exit 0; each dossier passed two-lane adversarial verification (independent harness audit plus mathematical audit) before inclusion here.

v1.8 session harnesses (five, written cold this session, one per dossier; every CL-, RD-, CH-, RN-, and PM- check ID cited in the v1.8 sections above appears in the corresponding file — the CL-2b(i)--(vii) row suffixes are assembled as the seven-row loop runs):

harness	content	exact + guards
<code>rd.softcheck.py</code>	OP-RADIUS: apex-ramification valuation certificate ($e = 2$ for r^2 , $e = 4$ for r) and the constructive-bridge obstruction; atom ledger with the cap-onset negative (Thm. 10.9–Rem. 10.11)	22 + 2
<code>cl.softcheck.py</code>	OP-RATE R6 classification no-go: extended k -invariance, the selection-functional no-go, the seven-way table, the Cencov-mirror closure (Lem. 2.16–Rem. 2.19)	20 + 5
<code>ch.softcheck.py</code>	χ doctrine g1/g3 resolved negatively: on-circle image of $S = \mu_2$, the charge character, the D2' dichotomy (Prop. 3.10, Rem. 3.11)	26 + 2
<code>rn.softcheck.py</code>	relational note: P4 recovered/reduced to P3, the bracketed odd floor, the bounded P1' transport-orbit scan (Rem. 4.9–Rem. 4.11)	20 + 4
<code>pm.softcheck.py</code>	period-relation map: τ_0 algebraic and no κ -vs- τ_0 relation, L_{res} as an elliptic period, the external-open ledger (Prop. 2.20, Rem. 2.21)	27 + 7
total		115 + 20

Across the thirteen session harnesses (`tx`, `mx`, `rx`, `ux`, `ex`, `cx`, `lx`, `dx`, `cl`, `rd`, `ch`, `rn`, `pm`) the running total is **349** exact checks + **49** certified guards, all exit 0. Environment pin: Python 3.12.10, sympy 1.14.0, mpmath 1.3.0 (the `py` launcher); every exact decision in $\mathbb{Q}/\mathbb{Q}(\sqrt{5})$ extended by i (and by $\sqrt{13} / 5^{1/4}$ where a Salem trace-down or an off-circle modulus enters), with $\mathbb{Q}(\sqrt{5})$ signs by rational arithmetic; the guards are interval-certified transcendental comparisons (mpmath `iv`, 60 dps), 50–140-dps quadrature/`pslq`/decimal corroborations, or fail-first falsifier rejections, none load-bearing; all five exit 0; each dossier passed two-lane adversarial verification, with the companion papers read from the Echo-S-Research archive read-only and their machinery re-encoded cold.

Every boxed identity of the source reports and every displayed identity of the review thread appears above with a passing check ID; the six findings (12.1, 12.2, 12.3, 12.4, 12.5, 12.6) are the complete defect list, unchanged in v1.8: the five v1.8 verifiers flagged only citation-hygiene in the derivation reports (companion-paper theorem numbers), corrected on the way in, and found no new display defect. The whitepaper's own harnesses (87+1 shipped; 89 independent) certify the upstream structure this paper conditions on.